

The Roof and the Clock

A verified computational model of $W^+(E_8)$, $O_8^+(2)$.3 and rowmotion on the E_7 minuscule poset, with three theorems on the non-linearity of rowmotion, and what the symmetry group and the clock share

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Status. Version 1.0, prepared on the Stenberg side from the sealed joint record for the second author’s final review; submission requires both authors’ word. The paper describes the spine both parties agreed on (SM-030, 038, 039, 040, 041, 047–053 of the joint registry) and the units added since (SM-054–066, §11), every one of which is JOINT: re-executed or re-derived independently by the second author and accepted on his word (Appendix A). The framing follows his publication response of 2026-09-08: classical facts are cited at first use, the computational model is presented as a model, the identification work as identification, and the one mechanism put forward as new is stated as theorems (§7.6) with the family of numbers behind it (§7.7) and a literature search behind the word “not found” (§7.4, Appendix B). Version history: v0.1 (2026-09-09, the spine); v0.2 (09-09, §§7.6–7.7, approved by the second author 09-10); v0.3 (09-11, §§7.8–7.13); v0.4–0.5 (09-13, §§11–12); v0.6 (09-13, §7.12.4); v0.7 (09-14, the second author’s acceptances and references); v0.8 (09-14, §11.9); v1.0 (09-19, stale status lines removed, Appendix B added, display formulas typeset; nothing computational changed or re-run since v0.3; §7.12.4’s B_8 paragraph added the same day, SM-067). Claims are graded: **[P]** classical with citation or proved here, **[C]** computed exactly by a named verifier under a brief locked before code, **[obs]** observed and not explained. No physical identification is asserted.

Abstract

We describe a computational model, built and verified in plain Python from first principles, of a chain of classical objects: the rotation subgroup $W^+(E_8)$ of the Weyl group of E_8 as the smallest group found to contain two towers of finite groups built earlier by the two authors separately; its quotient $\Omega = O_8^+(2)$; the triality automorphism realized as an explicit permutation Φ of order 3 on the 360 nonsingular points of the three 8-dimensional modules, so that $\langle \Omega, \Phi \rangle = O_8^+(2):3$ is an explicit permutation group; the fixed group of the normalized triality, $G_2(2)$, as an exact intersection of order 12,096; and the fourteen conjugacy classes of the twisted coset $\Omega \cdot \Phi$, enumerated with centralizers and power maps and matched class for class against the published character table of $O_8^+(2)$.3. On this frame we place a 56-state machine built independently on the Stenberg side, identify its 56 states with the vector block of the roof (the 56 weights of the minuscule representation of E_7), and show that its clock Ψ is exactly rowmotion on the 27-element E_7

minuscule poset, in the standard convention. The order of that rowmotion, 18, and its orbit structure [18, 18, 18, 2] are the Coxeter number of E_7 and the cycle type of a Coxeter element, as proved in general by Rush and Shi (2013); we do not claim them. What we add is the identification of a machine built by other means with that rowmotion, and a study of how far the Rush–Shi bijection is from an isometry. On the E_7 board a pair of states keeps its inner-product sign under one tick if and only if a parity computed from the two toggle sets is even, with zero exceptions on 1,485 pairs, and the clock commutes with no element of $W(E_7)$. Across the whole simply-laced minuscule family we prove that rowmotion is a Weyl element exactly on chains, that its half-turn is a Weyl element exactly on the vector representations of D_n , and that the fraction of pair types it keeps has closed forms on two families; the E_7 clock fails every linearity condition at the first place it can. Finally, the machine’s mirror pr is shown to be the E_6 diagram automorphism acting on the two 27-sheets’ ideal lattices, a classical object, and the clock and the mirror together generate the full symmetric group on the 56 states. Every registered guess that failed is reported at the same prominence as the results.

In §§7.8–7.13 we then ask what the symmetry group and the clock share, across the whole minuscule family. The Weyl group shares with any clock-translate of itself only the centralizer of that clock power, with the exceptions named and, by the end, explained; the longest element reverses the clock; three label counts are homomesic (Rush–Wang’s antichain homomesy read in the board’s letters) and so is the toggle size, a corollary of Defant–Hopkins–Poznanović–Propp; the one anomaly in the family has a mechanism — on the 4-cube the clock’s half-turn is a Weyl element on one of its two orbits — and, by the end, a reason: the element of that shape agrees with the half-turn on the middle-rank orbits of every spinor board with a free clock and commutes with it only on the 4-cube, a theorem for every even n (§7.12.4) — and the vector boards’ shared grammar is the centralizer of the clock at double speed; and the clock normalises no nontrivial subgroup of $W(E_7)$, so the Clifford-hierarchy analogy for it holds for one level and no further.

1. What this paper is, and is not

The paper has three kinds of content, and we mark which is which throughout.

1. **A verified computational model of classical structure.** The roof $W^+(E_8)$, the quotient $O_8^+(2)$, triality, the fixed group $G_2(2)$, the fourteen outer classes of $O_8^+(2)$.3, the E_7 minuscule poset and its rowmotion, the E_6 diagram automorphism: all of these are known objects [P]. Our contribution on them is a model that constructs each one explicitly from the previous, in one coordinate system, with every step checked by a script under a brief locked before the code was written, and the whole re-executed independently by the second author. §§3–6 and §8.
2. **Identification work.** Two purpose-built machines — a 56-state machine with three gates on the Stenberg side, a $PSL(2,7)$ tower on the Balashov side — are identified, object by object, with the classical structure: the 56 states with the vector block, the gate ι with a transvection, the clock with rowmotion, the mirror with the diagram

automorphism, each twisted class with its ATLAS name. This is real and narrow: it says what the machines are, not that the structure is new. §§6, 7.1, 9, 10.

1. **Three theorems and one rule on the non-linearity of rowmotion.** Rush–Shi make rowmotion on a minuscule poset conjugate to a Coxeter element; a Coxeter element is an isometry of the weights and rowmotion in general is not. We prove for which minuscule posets rowmotion, or its half-turn, is itself a Weyl element, give the defect in closed form on two families (§7.6), tabulate it for 42 cases (§7.7), and give the exact per-pair rule on the E_7 board (§7.2). A literature search — four targeted searches, the complete arXiv rowmotion corpus to September 2026, zbMATH Open, and the primary texts read for the question (§7.4, Appendix B) — found no match; we claim these as “not found”, not as new.

We do not claim the Coxeter-number periodicity of the clock (Rush–Shi), the structure of $O_8^+(2).3$ (ATLAS), the fixed group of triality (Conway, Kleidman), or the E_6 diagram automorphism’s action on minuscule posets (standard; see §10). Where earlier internal documents of ours used the phrase “the clock ticks at the Coxeter number” as a point of interest, this paper reads that fact as Rush–Shi’s theorem, seen on the data.

2. Background and conventions [P]

Weyl groups and lattices. $W(E_8)$ has order 696,729,600; its rotation subgroup $W^+(E_8)$ has index 2 and $W^+(E_8)/\{\pm 1\} \cong O_8^+(2)$, the simple orthogonal group of order 174,182,400 (Conway 1971; Conway–Sloane 1999, ch. 10; ATLAS). $W(E_7) \cong 2 \times \text{Sp}_6(2)$, and $\text{Sp}_6(2)$ acts on the 28 bitangents of a plane quartic (Dolgachev 2012, ch. 6). The 56 weights of the minuscule representation of E_7 are the roots r of E_8 with $\langle r, \alpha \rangle = 1$ for a fixed root α ; the E_7 root system is $\alpha^\perp$ (Bourbaki, ch. VI).

Triality and $G_2(2)$. $\text{Out}(O_8^+(2)) \cong S_3$; the fixed group of a triality automorphism of order 3 is $G_2(2) \cong U_3(3):2$ of order 12,096 (Conway 1971; Kleidman 1987; Wilson 2009, §4.7). The character table of $O_8^+(2).3$, with 53 classes of Ω fusing to 27 and 14 outer classes per coset, is in the ATLAS and in the GAP Character Table Library (Breuer, CTbLib 1.3.11).

Minuscule posets and rowmotion. For a minuscule weight λ of a simple Lie algebra, the weights of V_λ form a distributive lattice whose poset of join-irreducibles P_λ is the minuscule poset (Proctor 1984; Stembridge 1994; Green 2013); each join-irreducible is coloured by the simple root of its unique cover, and the multiplicity of colour i in P_λ is the coefficient of α_i in $\lambda - w_0\lambda$. Rowmotion on the order ideals $J(P)$ is $R(I) =$ the ideal generated by the minimal elements of $P \setminus I$ (Cameron–Fon-der-Flaass 1995; Striker–Williams 2012).

Theorem (Rush–Shi 2013). For P a minuscule poset, rowmotion on $J(P)$ is conjugate, under the natural bijection $J(P) \cong W\lambda$, to the action of any Coxeter element of W on the weights; in particular its order is the Coxeter number h and its orbit structure is that of a Coxeter element, and $(J(P), R, \text{rank generating function})$ exhibits the cyclic sieving phenomenon (Reiner–Stanton–White 2004). For E_7 , λ minuscule, $|P| = 27$, $|J(P)| = 56$, $h = 18$. Okada (2021) extends the statement to birational rowmotion. The simply-laced minuscule weights are

ω_k of A_n ; $\omega_1, \omega_{n-1}, \omega_n$ of D_n ; ω_1, ω_6 of E_6 ; ω_7 of E_7 . We normalize $\langle \alpha, \alpha \rangle = 2$, so that $\langle w, \alpha_i \rangle$ is the i -th Dynkin label of w .

Conventions of the model. All groups are permutation groups on explicit point sets (240 roots; 120 nonsingular vectors of each of the three 8-dimensional F_2 -modules; the 56 states), built from generators in one coordinate system and enumerated by Schreier–Sims where an order is claimed. Every numbered claim below names the verifier that computes it; the verifiers are in the registry’s envelopes together with their logs and their first-run logs.

3. The roof: $W^+(E_8)$ [C, SM-030]

Two towers were built earlier by the two authors: a chain of subgroups of $PSL(2,7)$ on the Balashov side ($PSL(2,7) \supset C_6 \times C_2, A_5, S_4, A_4, C_2$; developed, with its bridge into $Sp_6(2)$, in §11) and a tower of Schur covers and bridges on the Stenberg side ($SL(2,7), 2 \cdot I, GL(2,3), 2 \cdot O, 2 \cdot T, C_3 \times D_4, 2 \cdot W(E_6), 2 \times PGL(2,7)$, with the split covers Ih, Th). Stone V exhibits all sixteen members inside $W(E_8)$ with witnesses: the spine in a vector-type subgroup $C \cong Sp_6(2)$, the covers in $H \cong 2 \cdot Sp_6(2)$. The group $K = \langle C, H \rangle$ has order $348,364,800 = |W(E_8)|/2$, every generator has determinant $+1$ (exact Bareiss), so $K = W^+(E_8)$, the registered expectation; K is perfect and the 21 witness generators alone regenerate it, so the roof is minimal-found (not proven minimal — stated plainly). The two centres coincide, $z_H = -1$, so both towers hang over a single central C_2 , and $R/\langle -1 \rangle$ has the order of $O_8^+(2)$ [P, order only]. 44/44 checks.

Classical content: $W^+(E_8)/\pm \cong O_8^+(2)$ and the containment of $2 \cdot Sp_6(2)$ in $W(E_8)$ are known.
Computed: the explicit witnesses, the exact order, the perfectness, the single hinge.

4. The still point: $G_2(2)$ as an exact intersection [C, SM-038]

The triality of §2 is realized in the model as an outer automorphism τ' of the vector shadow $\bar{C} \cong O_8^+(2)$ (Stone X, SM-036, not in this paper). The intersection $\bar{C} \cap \tau'(\bar{C})$, computed by sifting all 1,451,520 elements of the vector shadow through the base-and-strong-generators of its image, has **order 12,096**, derived subgroup of order 6,048 and order census $\{1, 2, 3, 4, 6, 7, 8, 12\}$: it is $U_3(3):2 = G_2(2)$ [C + P]. The raw τ' does not normalize this group; an intertwiner y (unique up to the 2-dimensional solution space, one invertible Dickson-0 isometry) gives $\tau'' = \text{inn}(y^{-1}) \circ \tau'$ with **τ'' of exact order 3 fixing $G_2(2)$ pointwise**, and the triangle of the three shadows then closes on $G_2(2)$ — this is Kleidman’s fixed group of triality, seen as a set. Inside it, $PSL(2,7)$ occurs as **one class of 36 copies with normalizer $PGL(2,7)$** , all of the bridge class (the class transitive on the 28 bitangents). One registered expectation was INVERTED: the still point lifts split to $2 \times G_2(2)$ in the spin tower (no minus class among its involutions), where a twisted lift had been guessed. 19 PASS + 1 INVERTED.

5. The turn on states: $O_8^+(2):3$ explicit [C, SM-039]

Stone AA builds the odd half-spin module, finds its quadratic form Q^- unique and plus-type, and lets Ω act on the $360 = 3 \times 120$ nonsingular points of $V \oplus S^+ \oplus S^-$. The turn Φ is defined blockwise (T^{-1} on V , the unique intertwiner $S^+ \rightarrow S^-$, the closing map on S^-); it has **exact order 3**, conjugates the generators as τ does, and $|\langle \Omega, \Phi \rangle| = 522,547,200 = 3 \cdot |O_8^+(2)|$: $O_8^+(2):3$ as an explicit permutation group on 360 points. The still point is the pointwise stabilizer of the Φ -orbit $\{v, \Phi v, \Phi^2 v\}$ — one vector, one spinor, one co-spinor. 16/16.

Two measurements of this stone need the reframing of §1. Composing the turn with a bridge element g_7 of order 7 gives an element of order 21; composing it with the image of a Coxeter element of $W(E_8)$ gives an element of order **18**. Our internal notes called 18 “ E_7 ’s clock number at the roof”. Read against Rush–Shi, 18 is the Coxeter number of E_7 , and its appearance as the order of a twisted element is a fact about $O_8^+(2):3$ (the class 18A/B of §9), not a coincidence to be explained. What remains computed and ours is the explicit element and its class (§8).

6. The board under the still point [C, SM-040]

The 56-state machine (Stenberg side; states, edges and three gates Ψ , ι , pr sealed in `scar56_data.json`) is placed on the roof by one identification: **the 56 states are the 56 nonsingular vectors u with $B(u, v) = 1$** , equivalently the 56 E_8 roots with $\langle r, \alpha \rangle = 1$, equivalently the 56 weights of the E_7 minuscule representation — matched bijectively by Dynkin label, with all 84 crystal edges equal to subtraction of a simple root. Under this identification **the gate ι is the transvection $t_v: u \mapsto u + v$** (Dickson invariant 1, outside Ω), while Ψ and pr admit no linear extension. The still point is transitive on the 56 states, the 28 bitangents, the 36 even theta characteristics and the 63 Paulis (stabilizers 216, 432, 336, 192); each even theta’s stabilizer is a $PGL(2, 7)$ whose derived group is one of the 36 sevens of §4, bijectively. 14/14.

One measurement of this stone is the seed of §7: the fraction of state pairs whose incidence type is preserved is 100 % for ι and for every Weyl element, **93.0 % for pr , 65.1 % for Ψ** , 61.6 % for Ψ^9 , 54.7 % for Ψ^2 , 51.3 % for Ψ^3 , against a random mean of 48.2 % [obs].

7. The clock is rowmotion; the parity rule; the non-linearity theorems

7.1 The identification [C, SM-041]

The join-irreducibles of the sealed 56-state lattice form the 27-element E_7 minuscule poset P ; its 56 ideals are the 56 states; and **the sealed clock Ψ equals rowmotion $R(I) = \downarrow \min(P \setminus I)$ exactly** — not its inverse, not a conjugate; the convention is pinned on the data. This is the identification; by Rush–Shi it implies at once that Ψ has order 18 with orbit structure

[18, 18, 18, 2], which is what the machine had shown since its construction. In Stone AF (SM-044, not in this paper) the Coxeter elements of $W(E_7)$ were enumerated on the same 56 points: all 64 have cycle type [18, 18, 18, 2], Rush–Shi’s conjugacy seen on the data, and Ψ is not itself a Weyl element (best pointwise agreement 14 of 56). Stone AQ (§7.7) re-derives the same numbers from the Cartan matrix of E_7 alone, without the sealed data — an independent second derivation of the identification.

7.2 The rule [C, SM-041]

Write $\delta(u) = u + \Psi u$; δ is the XOR of the simple-root masks toggled by rowmotion at u (toggle sizes {1: 8, 2: 28, 3: 11, 4: 2, 5: 3, 6: 2, 9: 1, 27: 1}). For a non-antipodal pair $\{u, u'\}$, with t_1 the number of toggled colours of u at which u' has an edge, t_2 the same with the roles exchanged, and t_3 the number of Dynkin-adjacent pairs between the two toggle multisets:

$$\text{type}(u, u') \text{ is kept under } \Psi \Leftrightarrow t_1 + t_2 + t_3 \equiv 0 \pmod{2}.$$

Zero exceptions on all 1,485 non-antipodal pairs; with the 28 antipodal pairs the kept count is 1,002 of 1,540 = 65.06 %, the number of §6. The same rule with the k -fold toggle multiset is exact for every power Ψ^k ($k = 1..17$) and reproduces the whole measured profile. The eight cells (t_1, t_2, t_3) and their populations are tabulated in the stone; the dominant kept cell is (1, 1, 0) with 553 pairs.

7.3 Why it holds [P]

B is bilinear and the type of a non-antipodal pair is $B(u, u')$. Hence

$$B(\Psi u, \Psi u') - B(u, u') = B(\delta u, u') + B(u, \delta u') + B(\delta u, \delta u').$$

With δ the sum of the toggled simple roots β_c , $B(\beta_c, w') = \langle \alpha_c, w' \rangle \pmod{2} = |\text{label}_c(w')|$ for minuscule labels in $\{-1, 0, 1\}$, and $B(\beta_c, \beta_{c'})$ is the Cartan entry mod 2, i.e. Dynkin adjacency. The three terms are t_1, t_2, t_3 . **The content of the rule is therefore the identification of §7.1 and the count of §7.2**, not the algebra; once Ψ is rowmotion the rule is a one-line consequence of the bilinear form. The right object for the novelty question is therefore the one §§7.6–7.7 study: how far the Rush–Shi bijection is from an isometry, for which minuscule posets it is one, and what the defect is.

7.4 Literature status

The question is whether the statements of §§7.2, 7.6 and 7.7 — for which minuscule posets rowmotion, or a power of it, is itself a Weyl group element; the per-pair parity rule; the defect κ — are stated or proved anywhere. Seven searches were made; Appendix B records them. In brief:

- Two targeted searches on the Balashov side (toggle groups, periodicity, homomesy, cyclic sieving; rowmotion against an ambient inner-product sign on minuscule weights) and one on the Stenberg side (rowmotion with quadratic or bilinear forms

mod 2; the E_7 56-ideal case and the Gosset graph) found the Coxeter-motion and birational literature (Okada 2021) and the toggle literature, and no per-pair statement and no study of when the Rush–Shi conjugacy is realized by a Weyl element.

- A fourth (Balashov, 2026-09-10) followed the full citation trail of Rush–Shi (2013) on zbMATH Open — thirty citing documents, all read — with the same result.
- On 2026-09-19 (Stenberg side) the complete arXiv corpus of papers mentioning rowmotion — 57 documents, 2014 to September 2026, listed in Appendix B — was enumerated through the arXiv API and every title and abstract read; the zbMATH Open API was queried for the same term; and the primary texts were read for the specific question: Rush–Shi (2013), Okada (2021), Hopkins (2020) and Thomas–Williams (2019). None states that rowmotion on a minuscule poset is, or is not, an element of the Weyl group; none characterises the posets on which it or its half-turn is; none mentions an inner product, an isometry, or a per-pair rule.

Three adjacent results are cited for the specialist reader’s benefit and are not the same statement. Okada (2021) introduces *Coxeter-motion*, a product of file toggles, and proves it conjugate to rowmotion in the (birational) toggle group — two different maps related by conjugacy, the same relation Rush–Shi establish between rowmotion and a Coxeter element acting on W/W_J ; neither paper asks when the conjugating element can be taken trivial, which is the content of Theorems 1–2. Marczinzik–Thomas–Yıldırım (2024) study a homological “Coxeter matrix” (built from the incidence algebra’s Cartan matrix) alongside rowmotion, proving a different commutation identity. Panyushev (2016) gives a bijection indexing lower ideals of weight posets by Weyl group elements, not an identity between rowmotion and one. The easy half of Theorem 1 — on a chain, rowmotion is the cyclic shift, hence a Coxeter element — is classical (Rush–Shi attribute the chain and product-of-chains cases to Stanley; Striker–Williams 2012, Thm 6.1).

We therefore claim §§7.2, 7.6, 7.7 as “not found”, not as new. The searches were made without MathSciNet access; a specialist reader who knows these as corollaries of Rush–Shi’s conjugacy or of the toggle description of rowmotion is asked for the reference. The same status is claimed for §§7.8–7.13.

7.5 The E_6 control and two inversions [C, SM-041]

Each 27-sheet of the board (colour 6 deleted) is $J(P_6)$ with $|P_6| = 16$; its rowmotion has order $12 = h(E_6)$ with orbits $[12, 12, 3]$ (Rush–Shi again, and the sealed values of the machine’s E_6 ancestor); its pair types are the Schläfli adjacency; and the same rule with the E_6 diagram is exact there. Two registered guesses failed and are reported as such: the eight antipodal steps of Ψ are **not** the eight singleton toggles (two different 8-sets), and the E_6 clock keeps **63.5 %**, less than the E_7 clock’s 65.1 % — smaller toggle sets do not mean fewer flips.

7.6 The non-linearity theorems [P; checked C, SM-056]

Let λ be a simply-laced minuscule weight with poset $P = P_\lambda$, Coxeter number h , rowmotion R carried to the weights $W\lambda$. For $\lambda = \omega_k$ write c_k for the multiplicity of colour k in P (the coefficient of α_k in $\lambda + \lambda^*$), s_j for the number of elements of P of rank j ($0 \leq j \leq h-2$), and

$$\sigma = (s_0, \dots, s_{h-2}, c_k),$$

the *extended rank sequence*, of length h .

Lemma. (a) $R(\lambda) = w_0\lambda$; the bottom $w_0\lambda = -\lambda^*$ has a single nonzero label, -1 at the dual node k^* , hence a single cover $p_0 = w_0\lambda + \alpha_{k^*}$, which is the unique minimal element of P , and $R(w_0\lambda) = p_0$. Consequently

$$\langle \lambda, w_0\lambda \rangle = \langle \lambda, \lambda \rangle - c_k, \quad \langle w_0\lambda, p_0 \rangle = \langle \lambda, \lambda \rangle - 1.$$

(b) The R -orbit of $w_0\lambda$ is the sequence of rank truncations $P_{<j}$; writing $w(j)$ for their weights,

$$\langle w(j), w(j+1) \rangle = \langle \lambda, \lambda \rangle - \sigma_j \quad \text{for all } j \text{ (indices mod } h).$$

Proof. (a) $\emptyset \mapsto$ the ideal generated by $\min P = \{p_0\}$; and $\langle \lambda, \lambda - w_0\lambda \rangle = \sum_i c_i \langle \omega_k, \alpha_i \rangle = c_k$. (b) The minimal elements of $P \setminus P_{<j}$ are the rank- j elements and generate $P_{<j+1}$; $w(j+1) - w(j)$ is the sum of their colours' simple roots, each added at a label -1 of $w(j)$, so the inner product drops by s_j ; the wrap-around step $\lambda \rightarrow w_0\lambda$ drops by c_k by (a); all weights have equal norm. ■

Theorem 1.

$$R \in W \Leftrightarrow P \text{ is a chain} \Leftrightarrow \lambda \in \{\omega_1, \omega_n\} \text{ of } A_n;$$

then R is the Coxeter element $s_1 \cdots s_n$, the cyclic shift of $e_1, \dots, e_{n+1}$.

Proof. If $R \in W$ it is an isometry, so by Lemma (a) on the pair $(\lambda, w_0\lambda) \mapsto (w_0\lambda, p_0)$, $\langle \lambda, \lambda \rangle - c_k = \langle \lambda, \lambda \rangle - 1$, i.e. $c_k = 1$. From the classification, $c_k = \min(k, n+1-k)$ for $A_n \omega_k$, 2 for $D_n \omega_1$, $\lfloor n/2 \rfloor$ for $D_n \omega_n$ ($n \geq 4$), 2 for $E_6 \omega_1$, 3 for $E_7 \omega_7$; so $c_k = 1$ only for A_n with $k \in \{1, n\}$, whose poset is a chain. Conversely on the chain the ideals are the initial segments and R is the cyclic shift, a permutation matrix in $W(A_n) = S_{n+1}$. ■

Theorem 2. Let h be even and $m = h/2$. Then

$$R^m \in W \Leftrightarrow P \text{ is a chain, or } \lambda \text{ is the vector representation of } D_n, \quad n \geq 3$$

(with $D_3 \omega_1 = A_3 \omega_2$ and, for D_4 , the triality images ω_3, ω_4). In the vector case, on the weights $\pm e_i$,

$$R^{n-1}(e_i) = -e_{n-i} \quad (1 \leq i \leq n-1), \quad R^{n-1}(e_n) = (-1)^{n-1}e_n,$$

a signed permutation with an even number of sign changes, in $W(D_n)$.

Proof. (Obstruction.) If $R^m \in W$, it is an isometry with $R^m(w(j)) = w(j + m)$, so by Lemma (b) $\sigma_{j+m} = \sigma_j$ for all j : σ is m -periodic, and since $\sigma_0 = s_0 = 1$, $\sigma_m = 1$ is necessary. For $A_n \omega_k$ with $1 < k < n$ (n odd, $m = (n + 1)/2$), P is the $k \times (n + 1 - k)$ rectangle and $s_m = \min(k, n + 1 - k) \geq 2$. For $D_n \omega_n$, $n \geq 5$ ($m = n - 1$), P is the shifted staircase $\{(i, j): 1 \leq i \leq j \leq n - 1\}$ with rank $i + j - 2$, and $s_{n-1} = \lfloor (n + 1)/2 \rfloor - 1 \geq 2$. For $E_6 \omega_1$ ($m = 6$) the ranks are 1,1,1,2,2,2,2,2,1,1,1 and $s_6 = 2$; for $E_7 \omega_7$ ($m = 9$) they are 1,1,1,1,2,2,2,2,3,2,2,2,2,1,1,1 and $s_9 = 2$. For $D_n \omega_1$ ($m = n - 1$), P is a chain of $n - 2$ above the antichain $\{e_n, -e_n\}$ above a chain of $n - 2$, with $c_1 = 2$, and $\sigma = (1^{n-2}, 2, 1^{n-2}, 2)$ is $(n - 1)$ -periodic; the triality images at $n = 4$ and the chains pass likewise. (Sufficiency.) On $D_n \omega_1$ the ideals are $\emptyset$, the prefixes of the lower chain, the three ideals adding $e_n, -e_n$, or both, and the extensions along the upper chain; rowmotion cycles $\emptyset \rightarrow \dots \rightarrow P$ in $2n - 2$ steps and swaps the two one-sided ideals. On weights:

$$-e_1 \rightarrow -e_2 \rightarrow \dots \rightarrow -e_{n-1} \rightarrow e_{n-1} \rightarrow \dots \rightarrow e_1 \rightarrow -e_1, \quad e_n \leftrightarrow -e_n.$$

Hence $R^{n-1}(\pm e_i) = \mp e_{n-i}$ for $i \leq n - 1$ and $R^{n-1}(e_n) = (-1)^{n-1}e_n$: the coordinate permutation $i \leftrightarrow n - i$ with n (n even) or $n - 1$ (n odd) sign changes, even either way. ■

Corollary. $R^j \in W$ implies σ is j -periodic; so the linear powers of rowmotion are among the periods of σ . For E_7 , σ has no period below 18: no power of the clock but the identity is a Weyl element (SM-054's $C_W(\Psi) \cap \langle \Psi \rangle = 1$, by hand).

Theorem 3. Let $\kappa(R)$ be the fraction of unordered pairs $\{w, w'\}$ with $\langle Rw, Rw' \rangle = \langle w, w' \rangle$. Then, for $n \geq 3$,

$$\kappa(D_n \omega_1) = 1 - \frac{2(n-1)}{n(2n-1)}, \quad \kappa(A_n \omega_2) = 1 - \frac{3n^2 - 9n + 4}{\binom{n+1}{2}}.$$

Proof. ($D_n \omega_1$) Types are 0 and -1 (the n antipodal pairs). By the cycle above, $\{e_n, -e_n\}$ is the only antipodal pair R keeps; the other $n - 1$ become non-antipodal and, R permuting the pairs, $n - 1$ non-antipodal pairs become antipodal: $2(n - 1)$ of $n(2n - 1)$ pairs change. ($A_n \omega_2$) Weights are 2-subsets $\{a < b\}$ of $[n + 1]$; a pair's type is $|S \cap T| \in \{0, 1\}$. The ideals $(x_1 \geq x_2)$ of the $2 \times (n - 1)$ rectangle correspond to $\{n - x_1, n + 1 - x_2\}$, and rowmotion on the rectangle gives

$$R\{a, b\} = \{a - 1, b - 1\} \quad (a \geq 2, b \geq a + 2),$$

$$R\{a, a + 1\} = \{a - 1, n + 1\} \quad (a \geq 2),$$

$$R\{1, b\} = \{b - 2, b - 1\} \quad (b \geq 3),$$

$$R\{1, 2\} = \{n, n + 1\}.$$

With ρ the rotation $i \mapsto i - 1 \pmod{n + 1}$, a Coxeter element and an isometry, $R = \rho \circ \varphi$ where φ fixes every 2-subset outside $T = \{\{a, a + 1\}: 2 \leq a \leq n\} \cup \{\{1, b\}: 2 \leq b \leq n + 1\}$ and permutes T by $\{a, a + 1\} \mapsto \{1, a\}$, $\{1, b\} \mapsto \{b - 1, b\}$ ($b \geq 3$), $\{1, 2\} \mapsto \{1, n + 1\}$; so $\kappa(R) = \kappa(\varphi)$. A pair $(S \in T, G \notin T)$ changes type iff $a + 1 \in G$ ($S = \{a, a + 1\}$), iff $b - 1 \in G$

($S = \{1, b\}$, $b \geq 3$), or iff exactly one of $2, n + 1$ lies in G ($S = \{1, 2\}$): $2(n - 2)^2 + 2(n - 3)$ pairs. Inside T , of the $(n - 2) + 2(n - 1) + \binom{n}{2}$ intersecting pairs exactly $4n - 5$ stay intersecting, so $(n - 1)(n - 2)$ pairs change. Total $3n^2 - 9n + 4$. ■

Every step of these proofs was re-checked on the data (SM-056, 10/10): the lemma on all 42 cases of §7.7; $c_k = 1 \Leftrightarrow \text{chain} \Leftrightarrow R \in W$; the principal orbit and the rank sequences; periodicity $\Leftrightarrow R^{\{h/2\}} \in W$ with exactly the predicted case list; the vector formula for $n = 3..9$; the explicit rowmotion on 2-subsets and both κ formulas exactly for $n = 3..9$. A registered guess that the analogous count on $A_n \omega_3$ is a degree-4 polynomial in n was confirmed to $n = 10$,

$$12 D_3(n) = 15n^4 - 110n^3 + 165n^2 + 314n - 432,$$

suggesting degree $2k - 2$ for ω_k [obs].

7.7 The defect across the minuscule family [C, SM-055]

One machine built from the Cartan matrix alone (weights in Dynkin labels, the lattice by $w \rightarrow w - \alpha_i$ at label 1, its join-irreducibles, rowmotion as the join of the minimal elements of the complement) computes for 42 cases — A_n , all ω_k , $n \leq 7$; D_n , ω_1 and both half-spins, $n = 4..7$; E_6 ; E_7 — the kept fraction κ , the random baseline $\kappa_0 = \sum p_i^2$, the best pointwise agreement $a(R)$ with a Coxeter element, the altitude $v(R)$ (Hamming distance to the nearest Weyl element, W enumerated as permutations of the weights where $|W| \leq 322,560$; E_7 from SM-016), and the linear centralizer $C_W(R)$.

case	$ W\lambda $	h	R cycle type	$\kappa(R)$	κ_0	$a(R)$	$v(R)$	$ C_W(R) $
$A_n \omega_1$, ω_n	$n+1$	$n+1$	$(n+1)$	1	1	$n+1$	0	$n+1$
$A_3 \omega_2 =$ $D_3 \omega_1$	6	4	$4 \cdot 2$	11/15	0.680	3	3	2
$A_4 \omega_2$	10	5	5^2	29/45	0.556	5	5	1
$A_5 \omega_2$	15	6	$6^2 \cdot 3$	71/105	0.510	6	9	1
$A_5 \omega_3$	20	6	$6^3 \cdot 2$	58/95	0.452	7	13	1
$A_6 \omega_2$	21	7	7^3	76/105	0.500	10	11	1
$A_6 \omega_3$	35	7	7^5	339/59	0.419	10	25	1
			5					
$A_7 \omega_2$	28	8	$8^3 \cdot 4$	145/18	0.506	15	13	1
			9					
$A_7 \omega_3$	56	8	8^7	431/77	0.405	12	44	1
			0					
$A_7 \omega_4$	70	8	$8^8 \cdot 4 \cdot 2$	417/80	0.380	12	56	1
			5					

case	$ W\lambda $	h	R cycle type	$\kappa(R)$	κ_0	$a(R)$	$v(R)$	$ C_W(R) $
$D_4 \omega_1 (= \omega_3, \omega_4)$	8	6	6·2	11/14	0.755	4	3	2
$D_5 \omega_1$	10	8	8·2	37/45	0.802	5	5	2
$D_5 \omega_5$	16	8	8^2	19/30	0.556	5	10	1
$D_6 \omega_1$	12	10	10·2	28/33	0.835	6	5	2
$D_6 \omega_6$	32	10	$10^3 \cdot 2$	79/124	0.469	9	19	1
$D_7 \omega_1$	14	12	12·2	79/91	0.858	7	7	2
$D_7 \omega_7$	64	12	$12^5 \cdot 4$	295/504	0.432	13	48	1
$E_6 \omega_1$	27	12	$12^2 \cdot 3$	223/351	0.527	9	15	1
$E_7 \omega_7$	56	18	$18^3 \cdot 2$	501/770	0.482	14	38	1

(Dual representations give equal rows.) Rush–Shi’s conjugacy holds on the data in all 42 cases. The E_7 row reproduces §§6–7.1 and SM-054 from the Cartan matrix alone. Two registered guesses were inverted and are the content of Theorem 2 and of an observation: the linear centralizer is not trivial on the vector representations (it is the half-turn), and κ does not increase with rank along every family (it does along $D_n \omega_1$ and $A_n \omega_2$, and falls along the half-spins). κ exceeds κ_0 in every non-chain case. E_6 and E_7 sit where the D_5 – D_7 spinors sit; the 56-board’s clock is as far from linear as rowmotion on a minuscule poset gets.

7.8 The shared grammar on the 56 [C, SM-057]

7.8.1 Setting

$W = W(E_7)$, order 2,903,040, acting on the 56 states of the board (the weights of the minuscule representation). W is the board’s *grammar*: every symmetry, every Weyl-invariant selection rule, every descent marker (the involutions of a bridge $PSL(2,7)$), and the mirror up to one bit, lie in it. The clock Ψ is rowmotion on the E_7 minuscule poset (SM-041); it is not in W (SM-044) and commutes with nothing in W (SM-054). After k ticks the grammar is the conjugate $W_k = \Psi^k W \Psi^{-k}$, an isomorphic copy of W inside S_{56} .

7.8.2 Result 1 (what two instants share)

$$W \cap \Psi^k W \Psi^{-k} = \{1\} \quad (k = 1, \dots, 8, 10, \dots, 17), \quad W \cap \Psi^9 W \Psi^{-9} = \{1, \iota\}.$$

Computed by conjugating every element of W by Ψ^k and testing linearity; for $k \neq 9$ the cheap necessary test (the conjugate still commutes with ι) already leaves only the identity. The half-turn’s ι is forced: Ψ^9 commutes with ι (SM-044), and ι is central in W . Consecutive

instants of the machine share no symmetry at all; the eighteen instants are eighteen copies of $W(E_7)$ in S_{56} meeting pairwise in the identity (in $\langle \iota \rangle$ at distance 9), all with one spectrum.

Corollaries checked directly: no reflection of W (0 of 63) is a symmetry after any tick $k = 1..17$; none of the 21 involutions of a bridge $PSL(2,7)$ — the C_2 's of Ilya's descent — is a symmetry at any later instant short of the full cycle. SM-054 said no marker is *fixed* by the clock; this says no marker is even a *symmetry* one tick later.

7.8.3 Result 2 (the clock is universal, the mirror is not)

$$\langle W, \Psi \rangle = A_{56}, \quad \langle W, pr \rangle = 2^{28} \rtimes Sp_6(2), \quad \langle W, \Psi, pr \rangle = S_{56}.$$

Ψ, ι and the reflections are even permutations of the 56, pr is odd. The clock with the grammar generates every even permutation of the states: relative to the symmetry group, the clock is a universal gate — the permutation-group form of “the Clifford group plus one non-Clifford gate is universal”. The mirror with the grammar generates a group of order $2^{28} \cdot |Sp_6(2)| = 2^{37} \cdot 3^4 \cdot 5 \cdot 7$, and its structure is explicit: pr fixes exactly the two poles, which form an antipodal pair; pr composed with the swap of the poles is a Weyl element (SM-016's altitude $v(pr) = 2$, completed); W is transitive on the 28 antipodal pairs with image $Sp_6(2)$ and kernel $\langle \iota \rangle$; so the normal closure of the pole swap in $\langle W, pr \rangle$ is the full group of 2^{28} pair-flips, meeting W in $\langle \iota \rangle$, and $\langle W, pr \rangle = (2^{28}) \cdot W$. The mirror is one Pauli-type flip times a Clifford-type element; it lives in the sign-flip group of the 28 bitangents and cannot leave it.

7.8.4 What it means

- The early resource theory of the machine — structure is free, time is expensive; Ψ -depth 2, W -depth 4 (SM-009..014) — is the shadow of a dichotomy that is now a theorem on the board: everything linear sits in one finite Clifford-type world, the mirror one bit outside it, and the clock is the single operation outside that world, universal with it, rewriting all of it at every step.
- What the clock preserves of the grammar is exactly what the earlier stones measured: the spectrum (Rush–Shi: the orbits and the period of a Coxeter element), and a parity's worth of the pairwise relations (SM-041: 65 %, by an exact rule). Nothing else: not one non-trivial symmetry.
- For the Scar side: a selection rule is a statement about one instant of the machine. A choice made in the grammar — a marker, a generation axis — does not exist as a symmetry at any other instant. If the clock is to carry such a choice, what it carries is not a symmetry but an orbit, and the orbit leaves the symmetry group at the first tick.

7.8.5 Not claimed

Nothing about other minuscule boards (E_6 's sheets, D_n , A_n): the same computation is possible there and is the next stone, with the registered guess that $W \cap R^k W R^{-k} =$

$C_W(R^k)$ in general (it fits every number here, the half-turn included). The isomorphism type of $\langle W, pr \rangle$ beyond “the 2^{28} flips extended by W with $\langle \iota \rangle$ amalgamated” is not claimed.

7.9 The shared grammar across the family, and the reversal theorem [C + P, SM-058, SM-059]

7.9.1 Setting

As in §7.8: a minuscule board $W\lambda$ with Weyl group W (the grammar), rowmotion R (the clock), Coxeter number h . For $k = 1..h-1$ let $I_k = W \cap R^k W R^{-k}$ be the symmetry two instants k ticks apart share, and $C_k = C_W(R^k)$ the centralizer of the k -th power of the clock. Always $C_k \subseteq I_k$. On E_7 (SM-057), I_k is trivial except $I_9 = \{1, \iota\}$.

7.9.2 The shared grammar across the family (SM-058)

Computed for the 42 boards of SM-055 with W enumerated in full.

The centralizer law. On every board where no power of R is a Weyl element — the middle representations ω_k ($1 < k < n$) of $A_4..A_7$, the half-spins of D_6 and D_7 , E_6 and E_7 — $I_k = C_W(R^k)$ as sets, for every k : the symmetry two instants share is exactly the symmetry that commutes with the time between them. This was registered as a guess for all boards and INVERTED as a universal law: it fails on the chains ($R \in W$, so $I_k = W$ while $C_W(R^k)$ is small), on the vector representations of D_n (the half-turn is a Weyl element by SM-056, and $|I_k|$ follows $\gcd(k, n-1)$: on D_7 , $12 / 72 / 384 / |W|$ for $\gcd 1 / 2 / 3 / 6$, against centralizers $2 / 12 / 16 / 384$), and on one board it did not predict, the D_5 half-spin.

The half-turn survivor is the longest element. On every rich board with h even exactly one non-trivial symmetry survives the half-turn, and it is w_0 — not “ -1 when $-1 \in W$ and nothing otherwise”, as the registered table had it. On D_6 ’s half-spins four survive ($w_0 = -1$ among them), on D_5 ’s eight.

The D_5 anomaly [obs]. On the D_5 half-spin (16 weights = the even sign patterns of five axes, $h = 8$, R two free 8-cycles): $I_3 = I_5 = 2$ and $I_4 = 8$ against centralizers 1 and 4. The extra survivor at lags 3 and 5 is the pure coordinate permutation $(e_1 e_3)(e_2 e_5)$; at the half-turn the shared grammar is a dihedral group of order 8 containing the centralizer’s Klein four-group $\{1, -(e_1 e_2 e_3 e_4), (e_2 \leftrightarrow e_3, -e_4, -e_5), \text{their product}\}$, with two signed 4-cycles on e_1, e_4, e_2, e_3 . Two subgroups of order 1,920 in S_{16} meeting beyond the identity is not chance. It is not a hidden F_2 -linearity either: no power of the D_5 clock is affine on F_2^4 (checked; the same for D_6, D_7). Open.

7.9.3 The reversal theorem (SM-059)

Theorem. On every simply-laced minuscule board the longest element w_0 , acting on weights as $-\sigma$ (σ the diagram automorphism, trivial when $-1 \in W$), reverses rowmotion:

$$w_0 R w_0 = R^{-1}.$$

Proof. The antipode $w \mapsto -w$ reverses the root order on $W\lambda$, and σ preserves it; so w_0 is an antiautomorphism of the weight lattice, hence of $J(P)$. Rowmotion sends I to the ideal generated by the minimal elements of $P \setminus I$; its inverse sends I to the ideal whose complement is generated upward by the maximal elements of I . An antiautomorphism exchanges ideals with filters and minimal with maximal elements, so it exchanges these two recipes. ■ (Checked on all 42 boards: $w_0 \in W$, $w_0(\text{top}) = \text{bottom}$, $w_0 R w_0 = R^{-1}$.)

Corollary 1. w_0 commutes with R^k iff $R^{2k} = 1$; for h even, w_0 is a half-turn survivor, the one the family always has.

Corollary 2 (the clock never descends to the shadow). On a self-dual board the antipode ι is a permutation of $W\lambda$, and R^k preserves the pair $\{w, -w\}$ iff $R^{2k} w = w$. So the antipodal pairs preserved at lag k number $\text{fix}(R^{2k})/2$, and R^k induces a map on the shadow $W\lambda/\pm$ only when $R^{2k} = 1$. On E_7 : 1 pair for $k = 1..8$ (the axis), all 28 at $k = 9$ — SM-044's measured column, re-derived. (Checked on the 12 self-dual boards.)

7.9.4 The 56-board: two reversals and the mirror (SM-059)

The node-7 coweight splits the 56 into two poles ($\pm 3/2$) and two 27-sheets ($\pm 1/2$). Each sheet carries its own clock Ψ_6 , rowmotion on the sheet (order 12, cycle type $[12, 12, 3]$ per sheet), and its own reversal $w_0(E_6) \in W(E_6) \subset W(E_7)$, acting on the sheet's E_6 -labels as $-\sigma$ with σ the E_6 diagram automorphism ($0 \leftrightarrow 5, 2 \leftrightarrow 4$). Then:

- $w_0(E_6)$ reverses Ψ_6 and does not reverse Ψ ; $\iota = w_0(E_7) = -1$ reverses both.
- **The reversers are unique** (registered guess, CONFIRMED by full enumeration): exactly one element of $W(E_7)$ reverses Ψ , namely ι ; exactly one element of $W(E_6)$ reverses Ψ_6 , namely $w_0(E_6)$.
- **The mirror is the product of the two reversals:**

$$\text{pr} = \iota \circ w_0(E_6) \circ t, \quad t = \text{the swap of the two poles},$$

exactly on all 56 states. Consequences, each previously a separate sealed observation: $\text{pr} \circ t = \iota \circ w_0(E_6)$ is the Weyl element SM-016 found at Hamming distance 2 from the mirror and could not name (a fixed-point-free involution of $W(E_7)$); pr commutes with the sheet clock (SM-053's inverted guess AO6a) because both its factors reverse it; pr 's colour map is the E_6 diagram automorphism (SM-053 AO6b) because that is σ ; and pr keeps exactly a transposition's 1432 of 1540 pair types (SM-054's AP7) because it is an isometry times one transposition.

7.9.5 What it says

- The three gates of the machine are now named exactly: Ψ is time, the one operation outside the symmetry group and universal with it (SM-057); ι is the reversal of time, unique, and it is the antipode — on this board reversing time is flipping every state's

chirality; pr is the ratio of the board's reversal to the sheets' reversal, corrected at the poles. Time costs; both reversals are free.

- The shadow of the board — the 28 antipodal pairs, the classical Clifford world where SM-015 found ι acting trivially — has no clock. The clock preserves one pair per tick and needs the sign the shadow forgets. The board can reverse time only because it carries that sign.
- Across the family, what two instants share is the symmetry that commutes with the time between them, wherever time is genuinely non-linear; the longest word always survives the half-turn because it is the one Weyl element that runs the clock backwards; and the D_5 spinor board — the sixteen pure spin- $\frac{1}{2}$ patterns, one chirality of $\text{Spin}(10)$ — is the one board in the family that shares more; the mechanism is given in §7.12.2 and the reason in §7.12.4.

7.9.6 Not claimed

No proof of the centralizer law (a computation on 21 boards); no explanation of the D_5 anomaly; nothing about non-simply-laced boards; nothing physical — “the 16 of $\text{Spin}(10)$ ” is named as a representation, under Rule 3.

7.10 The D_5 anomaly on the 4-cube, and homomesy of the three letters [C, SM-060, SM-061]

7.10.1 The D_5 anomaly, located (SM-060)

SM-058 found the one board in the simply-laced minuscule family where the symmetry two instants share, $I_k = W \cap R^k W R^{-k}$, exceeds the centralizer $C_W(R^k)$ without any power of rowmotion being a Weyl element: the D_5 half-spin, the sixteen even sign patterns on five axes, Coxeter number 8, the clock two free 8-cycles; $I_3 = I_5$ of order 2, I_4 of order 8, against 1, 1, 4.

Where it lives. The B_4 spinor board — all sixteen sign patterns on four axes, Weyl group the hyperoctahedral group $\text{Aut}(Q_4)$ of order 384 — has the same ten-element poset and, under dropping the fifth coordinate, the identical rowmotion; $W(B_4)$ sits inside $W(D_5)$. The half-turn anomaly is already there: $|I_4| = 8$ against $|C_4| = 4$, with the same four extra elements, two signed 4-cycles $e_1 \rightarrow -e_4 \rightarrow -e_2 \rightarrow e_3 \rightarrow e_1$ and its inverse, and two involutions pairing (e_1, e_4) with (e_2, e_3) . Only the lag-3 survivor needs the fifth axis: it is the coordinate permutation $(e_1 e_3)(e_2 e_5)$, and it is the one survivor that mixes the clock's two orbits; every half-turn survivor preserves them, and so does $w_0 = -(e_1 e_2 e_3 e_4)$, which reverses the clock.

What it is not. - Not a hidden F_2 -linearity: no power of the D_5 (or D_6, D_7) clock is an affine map of the even code F_2^{n-1} ; D_4 's half-turn is, being the linear half-turn of SM-056. - Not a near-miss of linearity: the altitude of R^4 against $W(D_5)$ is 8 of 16, of R^3 7; the correction from the nearest Weyl element moves half the board. - Not the overgroup mechanism. The B_3 spinor shows the same signature (shared grammar 16 against centralizer 8 at its half-turn) for a visible reason: the B_3 board is the D_4 half-spin, whose half-turn is a Weyl element

of the larger $W(D_4) \supset W(B_3)$. On D_5 nothing of the kind: the half-turn generates the whole alternating group A_{16} together with $W(D_5)$. (On D_6 's half-spin, by contrast, $\langle W, R^5 \rangle$ stays inside the group preserving the sixteen antipodal pairs, as Corollary 2 of SM-059 requires.)
 - Not specific to the 4-cube in the naive sense: the B_5 spinor is clean ($I_k = C_k$ for every k); the registered guess was inverted only by the B_3 case above.

What it is, so far. The half-turn's shared grammar on the 4-cube is a dihedral group of order 8 generated by the centralizer (a Klein four-group: w_0 , and $(e_2 \leftrightarrow e_3, -e_4, -e_5)$, and their product) and one signed 4-cycle; the half-turn acts on it by an automorphism fixing the Klein subgroup and permuting the four extras among themselves, without inverting them (the registered guess that it inverts them was wrong). The anomaly is a statement about sixteen corners of a four-dimensional cube, a group of order 384, and a clock of order 8 that visits eight corners and then the other eight. It has resisted four explanations and is open.

7.10.2 Homomesy of the three letters (SM-061)

A statistic on states is *homomesic* under the clock if its average is the same on every clock orbit (Propp–Roby 2015). Rush–Wang (2015, arXiv:1509.08047) proved that on every minuscule poset the order ideal cardinality and the antichain cardinality are homomesic under rowmotion, and that the per-colour ideal cardinalities are (file homomesy). On the weights the latter says that every rowmotion orbit has mean weight zero; seen on all 42 boards.

The dictionary. A state's label at node i is $+1$ iff its ideal has a maximal element of colour i , -1 iff the complement has a minimal element of colour i , 0 otherwise. So $\#(+1 \text{ labels}) = |\max(I)|$, the antichain cardinality; $\#(-1 \text{ labels}) = |\min(P \setminus I)| = |\max(R(I))|$, the antichain cardinality of the next state; $\#(0 \text{ labels}) = \text{rank} - \text{both}$.

The theorem, read on labels. On every simply-laced minuscule board the three ternary counts are homomesic under the clock, with orbit means

$$\#(+1) = \#(-1) = \frac{|P|}{h}, \quad \#(0) = r - \frac{2|P|}{h},$$

where $|P|$ is the number of join-irreducibles, h the Coxeter number, r the rank. E_7 : $3/2, 3/2, 4$. E_6 : $4/3, 4/3, 10/3$. The D_5 spinor: $5/4, 5/4, 5/2$. The equality of the $+1$ and -1 means is Defant–Hopkins' 0-mesy of $T^+ - T^-$ (2021, arXiv:2108.13227), which holds on every finite poset.

One registered guess, confirmed. The toggle size — the number of elements of P that change between a state and the next, $|I \Delta R(I)|$ — is homomesic on all 42 boards with mean $2|P|/h$. This may follow from known toggleability results; it is flagged, not claimed as new.

Literature (Balashov, 2026-09-10): it does follow, in one line. Defant–Hopkins–Poznanović–Propp, Theorem 3.20, gives the antichain cardinality as homomesic with mean $|P|/h$ on every minuscule poset; the toggle size splits as $\max(I) \cup \min(P \setminus I)$, and $|\min(P \setminus I)| =$

$|\max(R(I))|$, so toggle size = antichain-card(I) + antichain-card($R(I)$), and the closure properties they state (a homomesic statistic composed with the map, and sums of homomesic statistics, are homomesic with the evident means) give $2|P|/h$. It is presented here as a corollary of their theorem, not as a finding of ours; the computation on 42 boards is its check.

Negatives. On every non-chain board three natural statistics are NOT homomesic: the number of distinct colours toggled, the per-state kept count of SM-041 (how many partners a state keeps its pair type with under one tick), and the number of lattice elements below a state. On the 56, under the mirror pr the ternary counts are not homomesic (a state and its image differ), and under the sheet clock Ψ_6 they are homomesic on the two 27-sheets alone (Rush–Wang on each E_6 sheet; the two poles are fixed points and break the count if included).

The reading [I]. With +1 as future, -1 as past and 0 as present: over every full cycle of the clock the average amounts of future and past are equal, and the average amount of present is conserved — on every board, exactly. This is Rush–Wang’s antichain theorem in the board’s own three letters. Time does not conserve which node carries which letter; it conserves, on average, how many of each kind there are.

7.10.3 Not claimed

No explanation of the anomaly (it is given in §7.12); the toggle-size homomesy is Defant–Hopkins–Poznanović–Propp’s corollary, not a claim of ours; nothing about non-simply-laced boards beyond B_3 , B_4 , B_5 ; nothing physical — “the 16 of Spin(10)” names a representation.

7.11 The reverser and the grammar [C + P, SM-062]

7.11.1 What is classical

For a minuscule representation with highest weight λ , the longest element w_0 of W sends λ to the lowest weight and acts on the weights as $-\sigma$, where σ is the diagram automorphism carrying λ to the highest weight of the dual representation (trivial exactly when $-1 \in W$: among the minuscule types, D_n with n even and E_7). Since -1 is central in GL , conjugation by w_0 acts on W as σ does: $w_0 s_i w_0 = s_{\sigma(i)}$. In particular w_0 is central in W exactly when $-1 \in W$, and then $w_0 = -1$ is the antipode of the weights. $\text{Aut}(\text{PSL}(2, 7)) = \text{PGL}(2, 7)$, of order 336, the outer automorphism swapping the two classes of elements of order 7 and the two three-dimensional representations. $\text{PSL}(2, 7)$ acts 2-transitively on the projective line over F_7 (eight points), with $\text{PGL}(2, 7) \subset S_8$ normalising it and centralising nothing.

7.11.2 What was computed (SM-062)

(a) The reversers are a coset [P, one line; C on 42 boards]. If g and g' both reverse rowmotion then $g^{-1}g'$ commutes with it; so the set of Weyl elements reversing R is $w_0 \cdot C_W(R)$. Measured: on all 41 enumerated boards the number of reversers equals

$|C_W(R)|$; on the 56, over all 2,903,040 elements, exactly one reverser (ι) and a trivial centralizer. With SM-054 ($C_{\{W(E_7)\}}(\Psi) = 1$) the uniqueness of SM-059 is a corollary. The reverser is unique exactly on the rich boards ($C_W(R) = 1$): 22 of the enumerated boards and the 56; there are h reversers on chains and 2 on the D_n vector boards.

(b) What the reverser does to the symmetry group [P + C on 42]. $w_0 s_i w_0 = s_{\sigma(i)}$ on every board. The reverser is central, and equals the antipode, on exactly seven boards of the family — $D_4 \omega_1/\omega_3/\omega_4$, $D_6 \omega_1/\omega_5/\omega_6$, $E_7 \omega_7$ — and on every other board it is non-central and acts on the grammar by the diagram flip: $\Lambda^k \leftrightarrow \Lambda^{n+1-k}$ on A_n , the two half-spins on D_{odd} , $27 \leftrightarrow 27^-$ on E_6 . On the 56 both are present: ι fixes all seven simple reflections, while the sheet reversal $w_0(E_6)$ of SM-059 flips the six E_6 reflections ($0 \leftrightarrow 5$, $2 \leftrightarrow 4$) and conjugates the seventh to a non-simple reflection (registered guess, CONFIRMED).

(c) The bridge copy's normaliser [C; registered guess CONFIRMED exactly]. For the bridge $PSL(2,7) = P$ of SM-013/SM-054: $C_{\{W(E_7)\}}(P) = \langle \iota \rangle$ and $N_{\{W(E_7)\}}(P)$ has order 672, so $N/C \cong PGL(2,7)$. $W(E_7)$ contains every automorphism of P and no further symmetry of it. The outer coset holds 56 involutions — 28 of cycle type $2^{24}1^8$ on the 56 (the guess from $x \mapsto 1/x$ fixing ± 1 on the projective line) and 28 fixed-point-free — 112 elements of order 6 and 168 of order 8. $N = \langle \iota \rangle \times PGL(2,7)$, with a $PGL(2,7)$ complement inside $Sp_6(2) = W^+$ (the fixed-point-free one) and one outside (SM-023's lemma seen).

(d) $C \neq T$ on the grammar [C, forced by SM-054/SM-059 given (c)]. ι commutes with every element of P . Any c inducing the outer automorphism is a Weyl element, hence commutes with no power of the clock and reverses none of them; measured: $c\Psi c^{-1}$ agrees with Ψ at four states and with Ψ^{-1} at two. Inside the whole normaliser the only reverser of Ψ is ι and the only element commuting with Ψ is 1.

(e) What the flavour conjugation has to act on [C, SM-013 cited]. P acts on the 28 antipodal pairs with fixed points 28/4/1/0/0, whose character is $\chi_1 + 2\chi_6 + \chi_7 + \chi_8$; the ι -even half of the 56 is this representation and the ι -odd half is isomorphic to it. No χ_3 or χ_3^- occurs. The outer automorphism therefore acts on the group P and on no triplet of states.

(f) Two observations beyond the brief [obs]. On the five self-dual boards where $-1 \notin W$ ($A_3 \omega_2$, $A_5 \omega_3$, $A_7 \omega_4$, $D_5 \omega_1$, $D_7 \omega_1$) the antipode is a permutation of the weights that reverses the clock and is not a Weyl element: a non-linear reverser, beside the linear ones. On those boards the isometry group of the weight configuration is $W \times \langle -1 \rangle$, so the isometry criterion for W -membership overcounts by the antipode; the sealed stones use the criterion on E_7 alone, where SM-054 makes it a theorem (checked).

7.11.3 What is identified, and what is read

Identification (Tier 2): in the machine's names, ι is the chirality gate and Ψ the clock (SM-041, SM-044); the antipode $w \mapsto -w$ is what charge conjugation does to the weights of a self-conjugate representation. So on the 56 the chirality gate is at once the antipode, the unique linear reversal of the clock, and central in the grammar: the board's C and its T are one element, and that element is invisible to every subgroup of $W(E_7)$. On the 27, where the

representation is not self-conjugate, the reverser is instead the $27 \leftrightarrow 27$ flip acting on the grammar. Both are exact.

Reading (Tier 3, Rule 3): on the board there is one operation that conjugates every state and reverses time, and it does nothing to the flavour group; the operation that conjugates the flavour group — the outer automorphism of $\text{PSL}(2,7)$, which is what the Scar side calls charge conjugation there — is a different Weyl element, a symmetry of the board at no later instant (SM-057), with no triplet of states to act on. Read as a constraint on any board-realised version of the lepton model: the model must contain an operation trivial on flavour that reverses a clock. This is a statement about the objects named, not about physical C, T, or leptons.

7.11.4 Open

Which of the 336 outer elements is “the” conjugation is a choice the board does not make, as it made no choice of marker pair (SM-054). The same question on the 27, where the reverser is non-central, is not computed. Whether the D_5 half-spin’s anomaly (SM-058/060) has anything to do with its board being one of the non-central cases is not known.

7.12 The anomaly by hand, and the doubled clock [P + C, SM-063, SM-064]

7.12.1 What is classical

The spin representation of B_n is minuscule; its poset is the shifted staircase δ_n , its weights the 2^n sign patterns $(\pm 1/2)^n$, its Weyl group the hyperoctahedral group of signed permutations, which is also the symmetry group of the n -cube on those vertices. Rowmotion on $J(\delta_n)$ has order $2n = h(B_n)$ (Rush–Shi 2013, for minuscule posets in general; here that is the statement, not a reframing). The D_{n+1} half-spin has the same poset and the same rowmotion, with $W(B_n) \subset W(D_{n+1})$. On the D_n vector representation (weights $\pm e_i$, minuscule, $h = 2n - 2$) the poset is the double-tailed diamond and rowmotion is a $(2n-2)$ -cycle on $\pm e_1 \dots \pm e_{n-1}$ times the transposition of $\pm e_n$.

$W(B_n)$ acting on the $2n$ points $\pm e_i$ is the centralizer in $\text{Sym}(2n)$ of the antipode $\iota = -1$, the stabilizer of the pairing $\{e_i, -e_i\}$; its conjugacy classes are the signed cycle types; a positive j -cycle of a signed permutation is two j -cycles on the $2n$ points and a negative j -cycle is one $2j$ -cycle, so the sign character of $\text{Sym}(2n)$ restricted to $W(B_n)$ is $(-1)^{\{\text{number of negative cycles}\}}$, and $W(D_n) = W(B_n) \rtimes \text{Alt}(2n)$. $W(B_4)$ acts on the 4-cube’s vertices, read as F_2^4 , as the affine maps $x \mapsto \pi x + v$ with π a coordinate permutation. A dihedral group of order $2m$ has two conjugacy classes of reflections when m is even and one when m is odd; when m is even the outer automorphisms that swap the two classes are not inner.

7.12.2 What was computed (SM-063, SM-064)

(a) The 4-cube’s half-turn is a Weyl element on one orbit [P + C]. Notation from §7.9: $I_k = \{g \in W : R^{\{-k\}gR}k \in W\}$ the grammar two instants share, $C_k = C_{W(R^k)}$, and the *transport* g

$\mapsto R^{\{-k\}gR}k$ on I_k . SM-058 found the one exception in the family to $I_k = C_k$ off the chains and vectors: the D_5 half-spin, located by SM-060 on the B_4 spinor (the 4-cube, W of order 384, R of order 8 with two free orbits $O_1 \ni \lambda$ and O_2), with $|I_4| = 8$ against $|C_4| = 4$. The mechanism, derived by hand and then checked (AY3–AY8):

- R^4 coincides on the whole of O_2 with $-\tau$, $\tau = (e_2e_3) \cdot (-e_4)$, an element of C_4 ; on O_1 it coincides with no Weyl element (the nearest agrees on at most 4 of its 8 points).
- O_2 affinely spans F_2^4 , so its pointwise stabilizer in W is trivial: a Weyl element is determined by what it does on O_2 .
- **Lemma B.** If $w \in W$ and $Y = \text{Fix}(R^k w^{-1})$ has trivial pointwise stabilizer in W , then $I_k \cap \text{Stab}_W(Y) = C_W(R^k w^{-1})$ and there the transport is conjugation by w . (Proof: for $g \in I_k$ preserving Y , $c^{-1}gc \in W$ agrees with g on Y .)
- Hence $I_4 = C_W(c)$ for the correction $c = (-\tau)R^4$, an involution that is the identity on O_2 and fixed-point-free on O_1 : eight elements, the four extras being those that commute with c but not with τ . The transport on I_4 is conjugation by τ , which explains SM-060's recorded permutation of the extras (the two reflections swap, the signed 4-cycle and its inverse swap), and $[E1, R^4] = -1$: an extra commutes with the half-turn up to the antipode.
- By hand: $\text{Stab}_W(O_1)$ is sixteen affine maps whose image in S_4 is the dihedral stabilizer of the pairing $\{14|23\}$, kernel ± 1 ; I_4 is the preimage of the Klein four-group; sixteen Weyl elements swap the two orbits and none is in I_4 . In binary phases along each orbit, C_4 acts by translations $\{000, 111, 001, 110\}$ on O_1 and $\{000, 111, 011, 100\}$ on O_2 , and R^4 is the translation 100 on both — in C_4 's image on O_2 only.
- The D_5 lag-3 survivor $(e_1e_3)(e_2e_5)$ falls to the same lemma with the nearest Weyl element w_3 (unique at Hamming distance 7; its agreement set of nine points determines $W(D_5)$): it commutes with $R^3w_3^{-1}$, and its lag-5 partner is its transport. Registered guess, CONFIRMED.
- The anomaly sits on B_4 , where the reverser -1 is central: it is not tied to D_5 's non-central reverser (the question left in §7.11.4).

(b) The vector boards' grammar is the centralizer of the doubled clock $[P + C]$. Three lemmas, proved before code and checked exactly:

- **Lemma C.** If an involution ι centralizes W and reverses R , then $I_k \subseteq C_W(R^{\{2k\}})$. (The transport h is fixed by ι , so $R^{kgR}\{-k\} = R^{\{-k\}gR}k$.) Checked at every lag on the eleven boards with such an ι : the seven with $-1 \in W$ ($D_4 \times 3$, $D_6 \times 3$, B_4) and the five whose antipode is not a Weyl element ($A_3\omega_2$, $A_5\omega_3$, $A_7\omega_4$, $D_5\omega_1$, $D_7\omega_1$; §7.11).
- **Lemma D.** On the D_n vector boards ($n \geq 3$, $D_3 = A_3\omega_2$) it is an equality: $I_k = C_W(R^{\{2k\}})$. (For g commuting with $R^{\{2k\}}$ the transport commutes with ι , hence lies in $W(B_n)$, and has the sign of g on the $2n$ points, hence lies in $W(D_n)$.) Checked as sets on $D_3..D_7$ at every lag. SM-058's "gcd($k, n-1$) pattern" on the vectors is this theorem: $|I_k| = |C_W(R^{\{2k\}})|$, and the guess $I_k = C_k$ failed there exactly because $C_W(R^{\{2k\}}) \not\supseteq C_W(R^k)$ off the half-turn.

- **Lemma E.** A transport preserves the unsigned cycle type and the sign, so it changes a signed cycle type only by trading two negative j -cycles for one positive $2j$ -cycle. No element of $W(B_n)$ makes such a trade; and g is W -inner (its transport is $w^{-1}gw$ for some $w \in W$) if and only if the signed types agree — the split D_n classes never separate g from its transport (registered, CONFIRMED). Every non-inner count of SM-063 is reproduced to the element: D_3 2 of 4; D_5 4 of 8 at the odd lags and 18 of 32 at lags 2, 6; D_7 6 of 12 at the lags coprime to 6 and 252 of 384 at lags 3, 9; none on D_4 , D_6 .
- D_3 by hand: $I_1 = W(D_2)$, the Klein group of coordinates 1, 2; the transport swaps the reflection $s_{\{e_1 - e_2\}}$ and the double flip — a transposition of S_4 traded for a double transposition.
- At every lag coprime to $n - 1$ the grammar is dihedral of order $2(n-1)$ [obs, post-reveal]. The transport preserves its cyclic half. For n odd it is the outer automorphism swapping the two reflection classes, and the inner half is exactly the cyclic half; for n even there is one reflection class and the transport is conjugation by the half-turn $R^{h/2}$ (a Weyl element, §7.6) or by $w_0 R^{h/2}$. The reason is one line: on $C_W(R^{2k})$ the transport equals conjugation by any odd power of R^k , and $h/2 = n - 1$ is such a power exactly when n is even. A registered guess that the realising pair lies in $C_W(R^{2k})$ was INVERTED; the pair is $\{R^{h/2}, w_0 R^{h/2}\}$.

(c) The auditor's error, at full size. Stone AY's brief stated and "proved" a Lemma A: $I_k = \bigcup_{w \in W} C_W(R^k w^{-1})$. The proof read " $R^{\{-k\}gR}k \in W$ " as " $R^{\{-k\}gR}k = w^{-1}gw$ for some $w \in W$ ", which is only the converse. The inclusion $\supseteq$ holds; equality holds if and only if every transport is W -conjugate to its element. Measured: equality on B_4 and D_5 at every lag; failure on the B_3 spinor (8 of the 16 elements of I_3), on $A_3\omega_2$ and on the odd D_n vectors — the very elements of (b). The machine found the error (the first run hung on the impossible set cover it produced; that log is kept). Lemma B never used Lemma A and stands. Two further post-reveal guesses in Stone AY (that the non-inner transports are antipode-conjugation, or inner in $W(B_n)$) failed; Lemma E says why. Across the 98 anomalous (board, lag) pairs of SM-058's table the picture is a dichotomy: on 83 a single Weyl element realises the transport, on 15 some transports leave their Weyl class outright; never does a pair need two. The B_3 spinor's non-inner transports are inner in the overgroup $W(D_4) \ni R^3$ (SM-060); the vectors' are inner nowhere below $\text{Sym}(2n)$.

7.12.3 What is identified, and what is read

Identified (mathematics): the family's one exception has a mechanism and the family's vector boards have a law. On the 4-cube the clock's half-turn is a symmetry of the board on one of its two orbits, and one orbit is enough to know a symmetry by; the shared grammar is the Weyl centralizer of the correction that makes the half-turn a symmetry on the other orbit too. On the boards whose only reverser is the antipode the grammar two instants share is the grammar the clock's double keeps, and the clock carries it forward by an automorphism that a Weyl element can copy exactly when the half-turn is an odd step of the clock.

Read [I], marked: SM-061 said no marker survives a tick and only orbit averages do; SM-063 says that where a marker does survive the half-turn (the four extras), it is because the half-turn is, for half the board, one of the board's own symmetries. The exception is not a crack in the rule; it is the rule seen from inside one orbit.

7.12.4 Why the orbit without λ [P + C, SM-066]

The question left open above — *why* the half-turn is a Weyl element on the orbit not containing λ — has a proof, not a search. Grade the spinor weights by rank (number of minus signs); $W(B_4) = \mathbb{F}_2^4 \rtimes S_4$ acts affinely, $w(x) = \pi(x) \oplus v$. **Lemma (Weyl rank-shift).** For $w = (\pi, v)$, writing $u = \pi^{-1}(v)$, $\text{rank}(w(x)) = \text{wt}(\pi(x) \oplus v) = \text{wt}(x \oplus u)$, so the rank-shift is

$$\Delta_w(x) = \text{wt}(u) - 2|x \cap u|$$

— a function of the single vector u . **Theorem [SM-066].** R^4 agrees with no Weyl element on O_1 and with $-\tau$ on O_2 . On O_1 the half-turn shifts every rank by exactly ± 2 (O_1 is the full-height orbit, its ranks reading 0,1,1,2,2, 3,3,4, and R^4 pairs positions differing by 2); at λ the shift is $+2$, forcing $\text{wt}(u) = 2$, whereupon $|\Delta| = 2$ forces $\langle x, u \rangle = 0$ for all $x \in O_1$, and O_1 spans $\mathbb{F}_2^4$, so $u = 0$ — contradiction. On O_2 the shifts are ± 1 , realised by $u = 1110$, i.e. $-\tau$. The cause is λ : the top weight pins $\text{wt}(u)$ at rank 0, so only an orbit avoiding the extreme ranks can carry the half-turn as a symmetry. The argument is dimension-free ($\Delta_w(x) = \text{wt}(u) - 2|x \cap u|$ in every B_n): on the full-height orbit of B_n the half-turn shifts rank by $\pm n/2$, so for every $n \geq 4$ the extreme-weight orbit's half-turn is not a Weyl element, the $n > 4$ case reducing to the fact that the shift is exactly $\pm n/2$ (verified at B_4 and at B_8 , SM-067).

B_8 : the anomaly does not recur [C, SM-067]. The clock acts freely on the 2^n weights of the B_n spinor board, with orbits of size $h = 2n$, only when $2n \mid 2^n$, i.e. n a power of 2, so B_8 (256 weights, W of order 10,321,920, $h = 16$) is the next board on which the two-orbit mechanism could recur and the last one in reach. There the clock has sixteen free orbits of sixteen; the orbit of λ is the full-height orbit, its half-turn shift exactly ± 4 , so Lemma BB-B applies; exactly two orbits carry the half-turn as a Weyl element, both by the same element $-(e_2e_3)(e_4e_5)(e_6e_7)$ (negating $e_1 \dots e_7$, fixing e_8), the analogue of $-\tau$, which is also the unique nearest Weyl element to R^8 (agreement 52 of 256); each is preserved by -1 . Lemma B applies, but the correction $c = w^{-1}R^8$ is no longer an involution (cycle type $8^8 4^{12} 3^{12} 2^{28} 1^{52}$) and its Weyl centralizer is $\{\pm 1\}$: $I_8 = C_8 = \{\pm 1\}$, and $I_k = C_k$ at every lag on B_8 , on the D_9 half-spin (the same clock; $W(D_9)$, of order 92,897,280, enumerated in full as affine maps $x \mapsto \pi(x) \oplus v$ on the even code), and on B_6 and B_7 . Among the spinor boards $B_3 \dots B_8$ the shared grammar exceeds the centralizer only on B_4 and, by the overgroup $W(D_4)$, on B_3 ; the conjecture that it does so exactly when n is a power of 2 is refuted at $n = 8$. The two carrying orbits are exactly those whose ranks stay within one of the middle rank ($\{3,4,5\}$; on B_4 , O_2 's $\{1,2,3\}$), equivalently those on which every half-turn rank-shift is ± 1 — an observation on two boards, not a theorem [obs]. One orientation slip in the record is corrected in passing: $R(\lambda) = w_0\lambda$ (§7.6, Lemma (a)), so the rank list 0,1,1,2,2,3,3,4 of SM-066 is read in the

direction of R^{-1} ; nothing in Theorem BB depends on the direction. Nothing about B_{32} or beyond.

The criterion, and the 16-cube [P + C, SM-068, SM-069]. Why B_4 and not B_8 is one commutation. For $w \in W$ and the correction $c = w^{-1}R^m$ ($m = h/2$), $c^2 = 1$ iff $R^m w R^m = w^{-1}$ — for an involution w , iff w commutes with R^m [P]; and if w commutes with R^m then w, c and R^m pairwise commute, any two of “ g commutes with w / with R^m / with c ” imply the third, and the extras of I_m on $\text{Stab}_W(\text{Fix } c)$ are exactly $C_W(c) \setminus C_W(w)$ [P] (SM-063’s “commute with c but not with τ ” as a lemma). Computed on $B_3..B_8$: the shared grammar at the half-turn exceeds the centralizer exactly when the Weyl element nearest to the half-turn commutes with it — $-\tau$ does on B_4 ; on $B_5..B_8$ the nearest element, always of the shape $-(e_2e_3)(e_4e_5)...$, does not, its correction is not an involution, and $C_W(c) = \{\pm 1\}$; on B_3 no element of $W(B_3)$ agrees with R^3 on a spanning set, so the criterion is a marker only and the B_3 excess is the overgroup’s (§7.10.1). On B_{16} (65,536 weights, W not enumerated), computed at the orbit level: the clock is free (2,048 orbits of 32), the orbit of λ shifts by exactly ± 8 , **sixteen** orbits carry the half-turn as a Weyl element, all by the same $-(e_2e_3)...\dots(e_{14}e_{15})$ with e_{16} fixed, and they are exactly the orbits with ranks in $\{7, 8, 9\}$ (the band, on three boards: 1, 2, 16 carrying orbits [obs]); that element does not commute with R^{16} , $C_W(R^{16}) = \{\pm 1\}$, the stabilizer of its 1,900-point agreement set is $\langle -1, w \rangle$ and $C_W(c) = \{\pm 1\}$, so by Lemma B no half-turn survivor beyond ± 1 preserves the agreement set; $|I_{16}|$ itself is not decided.

Why four — the commutation theorem [P, SM-070]. For even $n \geq 4$ the element $w_n = -(e_2e_3)(e_4e_5)...\dots(e_{n-2}e_{n-1})$ commutes with the half-turn R^n if and only if $n = 4$. The proof rests on a one-line recursion for rowmotion on the shifted staircase in “excess” coordinates $d_k = c_k - k$ (c_k the plus positions), from which the orbit of λ is written in closed form for every n : $R^{\{2j\}}\lambda = [n] \setminus \{n, n-2, \dots, n-2j+2\}$, $R^{\{2j+1\}}\lambda = [n] \setminus \{n-1, \dots, n-2j+1\}$, ranks 0, $n, n-1, n-1, \dots, 1, 1$ around the clock — so the half-turn shifts every rank on the full-height orbit by exactly $\pm n/2$ for every even n , and the hypothesis of the rank-shift theorem above is now proved for all n rather than verified at 4, 8 and 16. On that orbit w_n acts by $i \mapsto 2 - i$ (even clock positions) and $i \mapsto -i$ (odd), both commuting with $i \mapsto i + n$; on a carrying orbit commutation is automatic ($wHx = x = Hwx$). At $n = 4$ those two orbits are the whole board, which is why $-\tau$ commutes with R^4 . For even $n \geq 6$ the point e_2 (one minus sign, at position 2) is a witness: the closed forms $R^n(e_2) = \{1, 2\} \cup \{5, 7, \dots, n-1\}$ and $R^n(\{3, n\}) = [n] \setminus \{4, 6, 7, 9, \dots, n-1\}$ (both by induction along the orbit in excess coordinates) give $2 \in w_n R^n e_2$ but $2 \notin R^n w_n e_2$, since $w_n e_2 = -\{3, n\}$. Together with the criterion this is the reason for the 4-cube: the mechanism is present on every free clock; its consequence needs a commutation that only the smallest board allows. Still open: that w_n is the nearest Weyl element for $n > 8$, and $C_W(R^n) = \{\pm 1\}$ in general.

7.12.5 Open

Boards without a W -central reverser (the E_6 27, the D_{odd} half-spins, $A_n \omega_k$ with $2k \neq n + 1$): Lemma C does not apply and nothing is claimed. Whether $C_W(R^{\{2k\}}) \subseteq I_k$ beyond

the vectors: it is strict on $A_5\omega_3$, $A_7\omega_4$ and the D_6 half-spins at the half-turn. Which group the vectors' outer transports are inner in, if any short of $\text{Sym}(2n)$.

7.13 The clock's Clifford hierarchy [P + C, SM-065]

7.13.1 What is classical

In quantum computation the Clifford hierarchy (Gottesman–Chuang 1999) is the chain of sets C_1 = the Pauli group, $C_{j+1} = \{U : UC_1U^{-1} \subseteq C_j\}$; C_2 is the Clifford group and the T-gate lies in $C_3 \setminus C_2$: it is not a Clifford gate, but it sends every Pauli into the Clifford group. In group theory, for a subgroup $H \leq G$ and an element $x \in G$, the largest subgroup of H normalised by x is the intersection of the conjugates of H under $\langle x \rangle$, and the core of H in G is the intersection over all of G .

7.13.2 What was computed (SM-065)

The definition. For a board (W, R) and a lag k , the hierarchy at lag k is the chain $P_1(k) = W$, $P_{j+1}(k) = \{g \in P_j : R^{(-k)g}k \in P_j\}$; so $P_2 = I_k$, the grammar two instants share (§7.9), is the set of Weyl elements the clock sends back into W — the shape of a hierarchy level with W in the Clifford role. The depth $d(k)$ is the first j with $P_{j+1} = P_j$; the core $P_\infty(k)$ is where the chain stops.

Lemma F [P].

$$P_j(k) = \bigcap_{i < j} R^{ik} W R^{-ik};$$

the grammar shared by j consecutive instants at stride k . Hence the core is the largest subgroup of W normalised by R^k , and contains the core of W in $\langle W, R^k \rangle$.

Lemma G [P]. $P_3(k) = P_2(k)$ if and only if $I_k = I_{-k}$, if and only if a reverser of the clock in W (§7.9: w_0 is one, and they form the coset $w_0 \cdot C_W(R)$) normalises I_k — because conjugating I_k by R^k and by any reverser both give I_{-k} .

The table [C, 44 boards, 278 (board, lag) pairs]. Both lemmas were checked on every pair. Depth 1 exactly where $I_k = W$ (the chains, and the D_n vectors at the half-turn). Depth 2 everywhere else — two instants already share what all instants share — **with exactly one exception, the D_5 half-spin at lags 3 and 5**, where the survivor of §7.9 is carried by the clock to a different element (§7.12), no reverser normalises I_3 , and the chain drops once more: $1920 \supset 2 \supset 1$. The cores: $C_W(R^k)$ on the rich boards (trivial, or ± 1 at the half-turn where $-1 \in W$); $C_W(R^{2k})$ on the D_n vectors (§7.12); the dihedral I_4 of order 8 on the 4-cube and the D_5 half-spins at the half-turn; on the 56, from SM-057 without re-running, trivial at every lag but the half-turn, where it is $\{1, i\}$. Every one of these was a registered guess, including the exception; all held.

Normality [C]. Wherever the core is neither W nor of order ≤ 2 (44 cases) it is not a normal subgroup of W . Where $\langle W, R \rangle$ could be enumerated (the D_3 vector, the D_4 vector, the B_3

spinor: A_6, A_8, A_8 of orders 360, 20,160, 20,160) the core of W in $\langle W, R \rangle$ is trivial, strictly below $P_\infty(1)$ (orders 4, 6, 1).

7.13.3 What is identified, and what is read

Identified (mathematics): the run of shared grammars along the clock is short. On every board but one, what two instants share is what all instants share; the exception is the one survivor that time reversal does not preserve. The largest part of the symmetry group that the clock carries along unchanged is, on the rich boards and on the 56, nothing (or the antipode alone).

Read [I], marked: if the clock is the T-gate of the analogy and W its Clifford group, then the Pauli group is the core, and on the 56 there is none. The clock is beyond every level of the hierarchy relative to $W(E_7)$: no subgroup of $W(E_7)$, and in particular no copy of $PSL(2,7)$, is preserved by Ψ . On the vectors and the 4-cube the analogy holds for exactly one level — the clock maps a Pauli-like group into W by an automorphism, outer when n is odd (§7.12) — and W is not that group's Clifford group in any stronger sense, since it does not normalise it. SM-057's sentence, a Clifford-like W with one non-Clifford Ψ , is exact in this sense and in no other.

7.13.4 Open

$\langle W, R \rangle$ on the boards where it could not be enumerated (it is A_{16} on the D_5 half-spin by SM-060 and A_{56} on the 56 by SM-057; the pattern “alternating” is an observation on five boards, not a claim). A hierarchy in the other direction, grading the powers of the clock against a fixed subgroup of W , is not defined here. Nothing about the clock as a gate on a Hilbert space.

8. The twisted coset: fourteen classes, exactly [C, SM-047–051]

The conjugacy classes of $O_8^+(2).3$ lying in the outer coset $\Omega \cdot \Phi$ are classical data (ATLAS; CTbLib). Our model re-derives them from the explicit permutation group of §5 by centralizer closures (Schreier generators over transversals of the relevant fibres), with per-element conjugacy tests on sampled elements, and finds:

order	classes	centralizer in Ω	density
24	2	8, 8	$1/8 + 1/8$
21	1	7	$1/7$
18	1	6	$1/6$
12	4	32, 4, 48, 96	$5/16$
9	1	18	$1/18$
6	3	192, 24, 48	$13/192$
3	2	12,096 (the turn), 216	$1/12,096 + 1/216$
			sum = 1 exactly

Fourteen classes whose densities sum to 1, so there is no fifteenth. Along the way: the turn's centralizer in Ω is exactly $G_2(2)$ (its class has 14,400 elements); the eighteen's centralizer is its own cube's group, order 6; the "heart" $x = e^6$ of the eighteen has centralizer of order 1,944 with centre of order 9; the still point holds the heart (56 of its order-3 elements) and the eighteen's cube (504 of its order-6 elements) but no eighteen; both twenty-fours are made from the still point's two order-8 classes, one each. Four registered guesses were INVERTED across these stones and each was settled exactly by a post-reveal computation: the order-12 sample contains a third cycle type the still point never produces, carrying a quarter of the coset; the still point's own order-12 type splits into two classes (1/48 and 1/96); the heart of the still point turns into a turn, while its other order-3 class turns into the roof's commonest three-beat; and the first census overshot 1 by exactly 1/12,096, the duplicate that forced the last correction.

Framing: this is a re-derivation of published data by an independent route, presented as a check on the model, not as new group theory. Its value is that every later identification (§9) is made against numbers the model produced before the table was consulted.

9. The names [P cited + C, SM-052]

Against the GAP Character Table Library's table of $O_8^+(2).3$ (Breuer, CTbLib 1.3.11, ctoorth2.tbl; headers quoted verbatim with their source and hash; no character values used): the outer class count 14 is forced by the published class numbers (53 for Ω , 55 for $\Omega.3$, a fusion fixing 14 and fusing 13 triples); the multiset of (order, centralizer order) of the published outer classes equals ours tripled; all sixteen power-map entries our fibres had recorded agree; three entries computed fresh agree. The heart is class 3D of $O_8^+(2)$; the E_8 and D_4 Coxeter hearts are the fused triple 3A/3B/3C; the involution fibres are 2A and 2E; the two twenty-fours are told apart by their squares exactly as the table tells them apart. 17/17.

10. The mirror is the E_6 diagram automorphism [C on a P fact, SM-053]

The node-6 coweight splits the 56 into two poles and two 27-sheets. The machine's third gate pr swaps the sheets and fixes the poles, commutes with ι and — our registered guess inverted — commutes with the sheet clock Ψ_6 . The reason is the post-reveal finding: **pr is an order-isomorphism between the two sheets' ideal lattices $J(P_6^{(0)}) \rightarrow J(P_6^{(1)})$** (324/324 comparable pairs each way, principal ideals to principal ideals) whose induced map on the join-irreducibles is the colour map $0 \leftrightarrow 5, 2 \leftrightarrow 4, 1$ and 3 fixed: **the E_6 Dynkin diagram automorphism**. That the diagram automorphism swaps the two 27's and induces an isomorphism of their minuscule posets is standard (Bourbaki; Green 2013; see also Green–Xu 2024 for the statement “up to diagram automorphisms”); what is ours is that a gate built by other means is that automorphism. The gates generate everything: $|\langle \Psi, \iota \rangle| = 36$,

$\langle \text{pr}, \iota \rangle = 4$, and $|\langle \Psi, \text{pr} \rangle| = \mathbf{56!}$ by Schreier–Sims, so $\langle \Psi, \iota, \text{pr} \rangle = S_{56}$; the Cayley diameter of that generating set is at least 157 by counting [P]. 11 PASS + 1 INVERTED.

11. The other tower: $\text{PSL}(2,7)$, the 28 bitangents, and the bridge into $\text{Sp}_6(2)$ [P/C, SM-001, SM-003, SM-013, SM-015, SM-021–024]

Section 3 placed two towers inside $W^+(E_8)$; §§4–10 developed the Stenberg-side machine on that frame. This section collects the Balashov-side tower — the $\text{PSL}(2,7)$ framework — and the classical geometry in which the two frameworks meet: the 28 bitangents of a genus-3 curve, inside $\text{Sp}_6(\mathbb{F}_2) \cong W(E_7)/\{\pm 1\}$. It is a collation, on the Stenberg side, of the bridge results already sealed in the joint registry (SM-001, SM-003, SM-013, SM-015, SM-021, SM-022, SM-024/028) and of the standalone note *The Bitangent Bridge* (SM-008), independently re-verified by the second author (§11.8), with his own contribution as §11.9.

11.1 The $\text{PSL}(2,7)$ framework [P/C, SM-001]

$\text{PSL}(2,7) \cong \text{GL}(3, \mathbb{F}_2)$, order 168, with six conjugacy classes 1A, 2A, 3A, 4A, 7A, 7B of sizes 1, 21, 56, 42, 24, 24. Its character table, derived from scratch by the Burnside–Dixon class-algebra method and verified exactly over $\mathbb{Q}(\sqrt{-7})$ (`verify_tsc_scarcat.py`, 48 checks; 35+ rows of the second author’s registry independently confirmed):

	1A	2A	3A	4A	7A	7B
χ_1	1	1	1	1	1	1
χ_3	3	−1	0	1	α	$\bar{\alpha}$
χ_3^-	3	−1	0	1	$\bar{\alpha}$	α
χ_6	6	2	0	0	−1	−1
χ_7	7	−1	1	−1	0	0
χ_8	8	0	−1	0	1	1

with $\alpha = (-1 + \sqrt{-7})/2$. The Balashov-side tower is the odd spine $\text{PSL}(2,7) \supset A_5 \supset A_4 \supset C_2$ with Schur covers $\text{SL}(2,7)$, $2 \cdot \text{I} = \text{SL}(2,5)$, $2 \cdot \text{T} = \text{SL}(2,3)$ (each a non-split central $\mathbb{Z}_2$ -extension) and the split covers $\text{Ih} = A_5 \times \mathbb{Z}_2$, $\text{Th} = A_4 \times \mathbb{Z}_2$, $\text{GL}(2,3) = 2 \cdot S_4$; it is closed at the top by $W(E_6) < \text{Sp}_6(2)$ at index 28 (SM-021, `verify_tower_b.py`, 22 checks) — the second of the two towers §3 hangs over the single central C_2 .

11.2 No meeting at E_6 ; the field-level shadow [P/C, SM-003]

Because $7 \nmid |W(E_6)| = 51840$ and $168/27 \notin \mathbb{Z}$, $\text{PSL}(2,7)$ embeds in no $W(E_6)$ -geometry and has no transitive action on 27: the two frameworks cannot meet at the 27. At field level, $\text{Gal}(\mathbb{Q}(\zeta_{21})/\mathbb{Q}) \cong C_6 \times C_2$ has quadratic subfields $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{21})$ (Gauss sums $g_3^2 = -3$, $g_7^2 = -7$, $(g_3 g_7)^2 = +21$); $\mathbb{Q}(\sqrt{-7})$ is the character field of $\text{PSL}(2,7)$ — the unique

irrationality in the table, and the CM field of the Klein quartic — while $\mathbb{Q}(\sqrt{-3})$ is the Eisenstein field of the E_6 side. The two are linearly disjoint: at field level the frameworks meet only in $\mathbb{Q}$ (`verify_tbr_bridge.py`).

11.3 The bitangent home [C, SM-003]

Realize $\mathrm{PSL}(2,7) = \mathrm{GL}(3, \mathbb{F}_2)$ on $V \oplus V^* \cong \mathbb{F}_2^6$ with the symplectic form

$$\omega((v, f), (w, h)) = f(w) + h(v).$$

The 64 quadratic refinements of ω split by Arf invariant into 36 even and 28 odd. Computed in `verify_tbr_bridge.py` (20 checks):

- $\mathrm{PSL}(2,7)$ fixes exactly one even refinement; its orbits on the 36 even refinements are **[1, 7, 7, 21]** — the fixed form, the Fano points, the Fano planes, the flags [obs]. The stabilizer of an even form in $\mathrm{Sp}_6(2)$ is $O_6^+(2) \cong S_8$ (order $8! = 40,320$), and $\mathrm{PSL}(2,7)$ sits inside it by its 8-point action on $P^1(\mathbb{F}_7)$.
- $\mathrm{PSL}(2,7)$ is transitive on the 28 odd refinements (the bitangents), with stabilizer $S_3 = N(\langle z_3 \rangle)$ and permutation character $\chi_1 \oplus 2\chi_6 \oplus \chi_7 \oplus \chi_8$. The stabilizer of an odd form in $\mathrm{Sp}_6(2)$ is $O_6^-(2) \cong W(E_6)$, order 51,840.

11.4 The bridge [C + P, SM-003, SM-024/028]

Inside $\mathrm{Sp}_6(2) \cong W(E_7)/\{\pm 1\}$: **$W(E_6)$ is the stabilizer of one bitangent, $\mathrm{PSL}(2,7)$ is transitive on all 28, and $\mathrm{PSL}(2,7) \cap \mathrm{Stab}(\text{bitangent}) = S_3 = N(\langle z_3 \rangle)$.** This S_3 is at once (i) the “universal stabilizer” of the Scar-Cat registry ($28 = 168/6$), (ii) the canonical stabilizer of the strata decomposition $168 = 31 + 62 + 75$, and (iii) the type group S_3 of the E_6 architecture’s typed algebra — the two frameworks are the two point-of-view subgroups of one bitangent configuration, and the bridge closes the second author’s open “Bridge” conjecture. The embedding is canonical up to a single outer flip: $C_{\{\mathrm{Sp}_6(2)\}}(\mathrm{PSL}(2,7)) = 1$ and $N_{\{\mathrm{Sp}_6(2)\}}(\mathrm{PSL}(2,7)) = \mathrm{PGL}(2,7)$ at index 2, the outer automorphism realized as the symplectic duality $X \leftrightarrow X^*$ (SM-024 by an exhaustive 1,451,520-element pass; SM-028 by the second author’s structural route — two engines, one answer).

11.5 The bridge lands on the board [C, SM-003, SM-013]

The map $z \mapsto N(\langle z \rangle)$ is exactly 2-to-1 from the 56 elements of class 3A onto the 28 bitangent stabilizers — the group-internal incarnation of the $\pm$ -doubling by which the 28 bitangents index the 56 weights of the E_7 minuscule representation. This is the same 56 the clock runs on (§86, 7.1): $\mathrm{PSL}(2,7)$ acts on it with orbits **[28, 28]**, the two bitangent sheets interchanged by the antipode ι , weight-stabilizer S_3 (SM-013, answering design question DQ-1; the registered single-orbit guess refuted). The Weyl-level content is $\mathbb{C}^{56} = 2(\chi_1 \oplus 2\chi_6 \oplus \chi_7 \oplus \chi_8)$, the untwisted lift unique because $\mathrm{PSL}(2,7)$ is perfect; the quark sectors χ_3/χ_3^- do not occur in the Weyl action, only in the Lie-group embeddings of §11.6.

11.6 Branchings of the 27 [C, SM-003]

Two canonical restrictions of the 27 follow from the two embeddings: trification $E_6 \supset SU(3)^3$ with $PSL(2,7)$ diagonal by Klein's χ_3 gives $27| = 3\chi_1 \oplus 3\chi_8$; and $PSL(2,7) \subset G_2 = \text{Aut}(\mathbb{O})$ with $27 = J_3(\mathbb{O})$ gives $27| = 6\chi_1 \oplus 3\chi_7$. The earlier conjecture " $27 \leftrightarrow 2\chi_8$ " is impossible, as dimension 16 appears in no restriction of the 27.

11.7 The Clifford reading and the chiral Fano pair [C + obs, SM-015]

The bridge transports into quantum terms. $Sp_6(2) = W(E_7)/\pm$ is the 3-qubit Clifford group modulo phases and Paulis, exhibited computationally from both sides in shared coordinates (`verify_stoneq_clifford.py`, 35 checks): the 28 bitangents are the 28 odd Pauli sign-functions, $W(E_6)$ the Clifford stabilizer of one, and $PSL(2,7)$ a Clifford subgroup transitive on all 28, its fingerprint $[28] / [1,7,7,21] / S_3$ landing intact. The clock Ψ has **no** Clifford shadow ($\iota\Psi = \Psi^{-1}$ against central ι ; it does not act on the 28 pairs) — the group-theoretic face of the non-linearity of §7.6. Separately, on the E_6 architecture's 7-channel syndrome, exactly 2 of the 30 Fano structures on 7 labels are invariant under the syndrome's motion group S_3 — a chiral pair, "one bit" (a chirality convention) from a full Fano labelling, compatible but never forced [obs] (`verify_fano_syndrome.py`). The v1 "two counter-posed 27s" intuition, refuted as a description of the E_6 crystal, is true one floor up as the E_6 -branching $27 \oplus 27 \oplus 1 \oplus 1$ of the 56, with the mirror gate pr the diagram automorphism (§10).

11.8 What is the second author's, what is classical, what is joint

The skeleton — bitangents, theta characteristics, $O_6^\pm \cong W(E_6) / S_8$, $Sp_6(2) \cong W(E_7)/\pm$ — is classical. The $PSL(2,7)$ framework, its representation-ring selection rules, and the registry under audit are the second author's; SM-001 records that audit (35+ rows independently confirmed, his own predictions audit credited). What the joint work adds is the identification — that two independently built frameworks each singled out, for internal reasons, the *same* S_3 at the same place in this classical geometry — together with the [obs] items (the even-orbit structure $[1,7,7,21]$, the chiral Fano pair, the resurrection reading of §11.7) and audit-grade reproducibility. This section is a Stenberg-side collation of the joint record; the second author's independent re-verification of §11.3 (2026-09-14, the 36/28 split, $[1,7,7,21]$, the single 28-orbit and the $\chi_1 + 2\chi_6 + \chi_7 + \chi_8$ character, rebuilt from scratch) and his own expansion (§11.9) are merged, and the bibliography is completed.

11.9 The sextet χ_6 as the Fano plane's deleted permutation module [P + C, second author]

The second author contributes an explicit geometric model of the 6-dimensional irreducible χ_6 : it is the **deleted permutation module on the 7 points of the Fano plane** ($7 = 1 \oplus 6$, the singlet the overall labelling). Realizing $PSL(2,7) = GL(3, \mathbb{F}_2)$ on the 7 nonzero vectors of $\mathbb{F}_2^3$, the generators A (order 2), B (order 3), AB (order 7) have traces 2, 0, -1 on the 6-dimensional sum-zero subspace — exactly χ_6 's character at the classes 2A, 3A, 7A/7B —

which by character theory is a complete proof that this module is χ_6 . The same fact is stated, independently, in Luhn–Nasri–Ramond (2007, §7). His construction was re-run on the Stenberg side, reproducing the trace match exactly.

This is the same Fano plane as §11.3. The two size-7 orbits of $\text{PSL}(2,7)$ on the 36 even refinements — the “Fano points” and “Fano planes” of §11.3 — both carry permutation character $1 \oplus \chi_6$ (verified: the fixed-point counts (7,3,1,1, 0,0) over the six classes agree with the point action’s on the nose), so each is a 7-element $\text{PSL}(2,7)$ -set realizing χ_6 as its deleted permutation module: the points and the lines of the one Fano plane, dual under the diagram automorphism. The second author’s χ_6 -model is thus the “Fano points” orbit of §11.3 exactly, and the “Fano planes” orbit is its dual — resolving the question of whether these are the same object or a look-alike in favour of the same.

He further builds the six independent quartic invariants of χ_6 from the Fano combinatorics (the 7 points, the 7 lines, and an antisymmetric pairing), rank 6 by an explicit computation — a representation-theoretic count. Their use in the King–Luhn flavon-potential and vacuum-alignment programme is a physical identification and is **not** taken up here (Rule 3).

12. Discussion

We collect here what the parts establish together. Nothing in this section is a new claim; each sentence points back to a graded result above.

One frame holds two machines. The organizing fact of §§3–6 is that a single classical group, the rotation subgroup $W^+(E_8)$, was found to contain two towers built independently and for unrelated reasons — a chain inside $\text{PSL}(2,7)$ on the Balashov side, a tower of Schur covers on the Stenberg side — over one central C_2 [C, SM-030]. Everything downstream lives on that frame in one coordinate system: the quotient $O_8^+(2)$, the triality Φ realized as an explicit order-3 permutation of the 360 points (§5), the still point $G_2(2)$ recovered as an exact intersection with $\text{PSL}(2,7)$ sitting inside it as the bitangent class (§4), and the fourteen outer classes of $O_8^+(2)$.3 enumerated and matched name-for-name to the published table before any character value was read (§§8–9). The value of the construction is not that these objects are new — they are not, and §14 says so — but that they are built, not asserted, each from the last, and re-executed independently by the second author (§15).

The clock on that frame is rowmotion. The 56-state machine, built by other means on the Stenberg side, has its states identified with the vector block of the roof — the 56 weights of the minuscule representation of E_7 — and its clock Ψ shown to be exactly rowmotion on the 27-element E_7 minuscule poset in the standard convention [C, SM-041]. That this rowmotion has order 18 with orbit structure [18, 18, 18, 2] is the Coxeter number and Coxeter-element cycle type, and is Rush–Shi’s theorem, not ours (§7.1). What is ours is the identification of a machine with that rowmotion, and the mirror gate pr with the E_6 diagram automorphism (§10), after which the two gates and the clock generate the full symmetric group on the 56 states.

The group and the clock share little, and exactly how little is the theorem. The question of §§7.8–7.13 — what a Weyl group shares with the translates of itself by its own clock — has a sharp answer: the intersection of $W(E_7)$ with the centralizer of a clock power is the centralizer and nothing more, meeting $W(E_7)$ itself in the identity alone [C, SM-054]; the longest element reverses the clock [C + P, SM-059]; three label counts and the toggle size are homomesic, the last as a corollary of Defant–Hopkins–Poznanović–Propp [C, SM-060/061]; the one anomaly in the family carries a mechanism rather than an exception (on the 4-cube the half-turn is a Weyl element on one orbit, §7.12); and the Clifford-hierarchy analogy for the clock holds for exactly one level and no further [P + C, SM-065]. Read together with the non-linearity theorems (§§7.6–7.7) — that rowmotion is a Weyl element exactly on chains, its half-turn exactly on the D_n vectors, with the defect in closed form on two families and tabulated for 42 cases — the picture is consistent: the E_7 clock is as far from being a symmetry of the board as it can be, and fails every linearity condition at the first place it can (§7.6). The parity rule (§7.2) is the local face of the same fact: a single tick preserves an inner-product sign iff a toggle-set parity is even, exactly, on all 1,485 pairs.

Method. Two disciplines carried the work and are worth stating as findings in their own right. Every unit was sealed under a brief whose SHA-256 was fixed before its script existed, so that each identification in §§9–10 is made against numbers the model produced before the reference table was consulted; and every registered expectation that failed is reported at the same prominence as those that held (§13), including the second author’s and our own — nineteen inversions and one lemma reported false as stated (§7.12) across the run. The model’s credibility rests on those two rules more than on any single tally.

Outlook. Several directions are open and are named here as future work, not as gaps in the present claims. The non-simply-laced minuscule weights ($B_n \omega_n, C_n \omega_1$), promotion, and birational rowmotion (Okada 2021) are untouched here; a closed form for the defect κ on the spinors, on E_6 and E_7 , and on $A_n \omega_k$ for $k \geq 3$ beyond the observed degree at $k = 3$ is not known to us; and the “not found” status of §§7.2, 7.6, 7.7 (§7.4, Appendix B) is exactly that: a specialist reader with MathSciNet access may know these as corollaries, and the reference would be welcome. Whether the coincidence of the number 18 between the board’s Coxeter number and the roof’s twisted classes (§§5, 8) is more than a coincidence is left as it was found — two different objects (§7.1) — and any reading of the clock beyond combinatorics remains marked [I] and unclaimed (§14).

13. Refutations, at equal prominence

Across the twelve spine units, the registered expectations that failed:

- SM-038: the still point lifts split, not twisted.
- SM-041: the antipodal steps are not the singleton toggles; the E_6 clock keeps less than the E_7 clock.
- SM-050: the order-12 twisted types are not the still point’s two; the still point’s type is two classes, not one.

- SM-051: the heart at the still point turns into a turn, not into the class first guessed.
- SM-053: the mirror commutes with the sheet clock rather than normalizing it into another power.

And in the two units of §§7.6–7.7: SM-055’s guesses that the linear centralizer is trivial off the chains (false on the vector representations, where it is the half-turn) and that κ rises with rank in every family (false on the half-spins). Each is recorded in its stone with the first-run log kept. Nine first runs across the September stones stopped on instrumentation errors of the executor; none touched a registered result; every such log is shipped.

In the nine units of §§7.8–7.13: SM-058’s guess that the shared grammar is the centralizer on every board (inverted on the chains, the D_n vectors and the D_5 half-spin — the last explained in §7.12, the vectors made a theorem there) and its predicted table; SM-060’s guesses that the anomaly is specific to the 4-cube (inverted on B_3) and that the half-turn inverts the extras; SM-061’s guess about the mirror and the sheet clock; SM-063’s guess that a single Weyl element realises the transport on every anomalous pair (inverted on 15 of 98) and its Lemma A, stated as proved and false as stated (§7.12.2(c)); SM-064’s guess on where the even- n realising pair lies; and five slips of the executor’s own (SM-060 twice, SM-062 once, SM-063 twice), each reported in its row. SM-057, SM-059 and SM-065 had every registered guess confirmed.

In SM-067 (§7.12.4, B_8): the guesses that the half-turn anomaly recurs on the 8-cube ($|I_8| > |C_8|$, ratio 2) and that it marks the boards with a free clock (inverted: $I_k = C_k$ at every lag on B_8 , the D_9 half-spin, B_6 and B_7), that -1 exchanges the two carrying orbits (it fixes each), and that D_9 carries a survivor at a lag coprime to 16 (none); two scoring slips of the executor’s own (the rank list read against the clock’s direction; a containment bar that demanded extras which do not exist), both kept in the first-run log. In SM-068 (the criterion): that the agreement set on B_8 has no Weyl symmetry beyond ± 1 (it has $\langle -1, w \rangle$), that the criterion explains B_3 from inside $W(B_3)$ (no spanning agreement set exists there), and that the nearest element is unique only up to the antipode (it is unique outright). In SM-069 (the 16-cube): that two orbits carry the half-turn (sixteen do, -1 swapping four pairs of them) and that every orbit spans (61 do not); one instrumentation error in the centralizer search, fixed and cross-checked, in the first-run log.

14. What is not claimed

No physical identification of any object here. No claim that the Coxeter-number periodicity, the fixed group of triality, the outer classes of $O_8^+(2).3$, or the action of the E_6 diagram automorphism on minuscule posets are new. No claim of novelty for §§7.2, 7.6, 7.7 beyond “not found” (§7.4, Appendix B). No claim that the roof is minimal beyond minimal-found. Nothing about the non-simply-laced minuscule weights ($B_n \omega_n, C_n \omega_1$), about non-minuscule posets, promotion, or birational rowmotion. No formula for κ on the spinors, on E_6, E_7 , or on $A_n \omega_k$ with $k \geq 3$ beyond the observed polynomial degree at $k = 3$.

No claim of novelty for §§7.8–7.13 beyond “not found” (§7.4, Appendix B); the toggle-size homomesy of §7.10 is Defant–Hopkins–Poznanović–Propp’s, not ours. “Clifford”, “Pauli” and “T-gate” in §7.13 are labels for a chain of subgroups; no gate on any Hilbert space is claimed. Stone AY’s Lemma A (§7.12) is false as stated and is used nowhere. Nothing physical: no lepton, no flavour model, and no reading of the clock as time beyond the paragraphs marked [I].

15. Reproducibility

Every unit is a Python 3 script with numpy (and sympy in one place), run under a brief whose SHA-256 was recorded before the script existed; each script re-checks the brief’s hash as its first test. The second author re-executed all twelve spine units from clean checkouts on an independent machine (2026-09-08): every tally matched, and the logs of eight units were byte-identical modulo line endings and timings. His audit found that several verifiers depend on caches shipped in earlier envelopes and not named in the cover notes; the dependency table below is the correction:

unit	verifier	external inputs
SM-030	verify_stone_v_we8_roof.py	_stone_u_cache/ (Stone U, regenerable by its own verifier)
SM-038	verify_stone_z_stillpoint.py	_stone_u_cache/, _stone_v_cache/, _stone_w_cache/ts_subspaces.npz (regenerable by Stone W’s verifier), _stone_x_cache/
SM-039	verify_stone_aa_roofclock.py	the above + _stone_z_cache/
SM-040	verify_stone_ab_merkabit.py	scar56_data.json, _stone_z_cache/G_rows.npy
SM-041	verify_stone_ac_parity_rule.py	scar56_data.json only
SM-047–051	verify_stone_ai..am_*.py	Stone AA’s caches (_stone_aa_cache/), _stone_z_cache/, the Stone U/V/X caches, each replayed from scratch at the head of the script
SM-052	verify_stone_an_atlas_names.py	the above + the two cited CTblLib header files shipped

unit	verifier	external inputs
SM-053	verify_stone_ao_mirror_and_gates.py	in _stone_an_cache/ scar56_data.json only
SM-055	verify_stone_aq_rush_shi_defect.py	none (Cartan matrices only)
SM-056	verify_stone_ar_nonlinearity_theorems.py	the file verify_stone_aq_rush_shi_defect.py (its machine is loaded verbatim; hash logged)
SM-057	verify_stone_as_shared_grammar.py	SM-054's verifier file and cache _stone_ap_cache/witnesses_ap.json, scar56_data.json
SM-058	verify_stone_at_grammar_family.py	the AQ verifier file (machine loaded verbatim)
SM-059	verify_stone_au_two_reversals.py	SM-055's and SM-054's verifier files, scar56_data.json
SM-060	verify_stone_av_d5_anomaly.py	none (boards built in e-coordinates)
SM-061	verify_stone_aw_homomesy.py	the AQ verifier file
SM-062	verify_stone_ax_c_equals_t.py	the AQ, AT, AP, AS and AU verifier files, scar56_data.json, _stone_ap_cache/witnesses_ap.json
SM-063	verify_stone_ay_d5_by_hand.py	the AV, AQ and AT verifier files, _stone_at_cache/table_at.json
SM-064	verify_stone_az_doubled_clock.py	the AV, AQ and AT verifier files, _stone_at_cache/table_at.json
SM-065	verify_stone_ba_clifford_hierarchy.py	the AQ, AT and AV verifier files

The kbar_* files of Stone U (~180 MB) are excluded from every envelope and are regenerated by that stone's verifier.

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Appendix A. The registry units and their seals

Spine: SM-030 (Stone V, 44/44), SM-038 (Stone Z, 19 + 1 inverted), SM-039 (Stone AA, 16/16), SM-040 (Stone AB, 14/14), SM-041 (Stone AC, 9 + 2 inverted), SM-047 (Stone AI,

15/15), SM-048 (Stone AJ, 17/17), SM-049 (Stone AK, 14/14), SM-050 (Stone AL, 15 + 2 inverted + 1 post-reveal), SM-051 (Stone AM, 15 + 1 inverted + 1 post-reveal), SM-052 (Stone AN, 17/17), SM-053 (Stone AO, 11 + 1 inverted + 1 post-reveal). All twelve JOINT on both parties' words as of 2026-09-06; all twelve re-executed on the Balashov side 2026-09-08 with matching tallies.

Added in v0.2 (§§7.6–7.7): SM-055 (Stone AQ, 8 + 2 inverted, 2026-09-09) and SM-056 (Stone AR, 10/10, 2026-09-09) — JOINT on the second author's word of 2026-09-10 (re-run and accepted on his side). Related: SM-054 (Stone AP, 22 + 2 disclosed, 2026-09-07) — the centralizer of Ψ in S_{56} meets $W(E_7)$ in the identity alone — JOINT on his word of 2026-09-10, together with SM-035, SM-036 and SM-037 of the earlier lane.

Added in v0.3 (§§7.8–7.13), **JOINT on the second author's word of 2026-09-14** (he read all of §§7.8–7.13 and accepted each): SM-057 (Stone AS, 14/14), SM-058 (Stone AT, 7 + 4 inverted or corrected), SM-059 (Stone AU, 10/10), SM-060 (Stone AV, 9 + 4), SM-061 (Stone AW, 6 + 1), SM-062 (Stone AX, 21 + 1), SM-063 (Stone AY, 20 + 5, including its own Lemma A reported false), SM-064 (Stone AZ, 11 + 2); SM-065 (Stone BA, 7/7, 2026-09-11) — JOINT on his word of 2026-09-14.

Added in v0.6 (§7.12.4): SM-066 (Stone BB, 6/6, 2026-09-13) — the half-turn on the full-height orbit, closing SM-063 §7.12.4 open #1 by the Weyl rank-shift lemma — JOINT on the second author's word of 2026-09-14 (Lemma BB-A rebuilt from scratch on his side: $W(B_4) = \mathbb{F}_2^4 \rtimes S_4$ of order 384, the rank identity on 200 random triples, the contradiction by hand).

Added in v0.8 (§11.9): the second author's model of χ_6 as the Fano plane's deleted permutation module, his construction re-run on the Stenberg side; the identity with §11.3's size-7 orbits verified (`verify_fano_bitangent_identity.py`). All thirty-six units cited above are JOINT as of 2026-09-14.

Added 2026-09-19 (§7.12.4, the B_8 paragraphs): SM-067 (Stone BC, 14 + 4 inverted), SM-068 (Stone BD, 7 + 3), SM-069 (Stone BE, 9 + 2), SM-070 (Stone BF, 8/8) — the 8-cube, the criterion, the 16-cube and the commutation theorem, prepared on the Stenberg side under briefs locked before code; the second author's re-execution is invited. They are the four units cited in this paper that are not yet JOINT.

Appendix B. The literature search behind “not found”

The claim of §7.4 is negative and is therefore only as good as the search behind it. This appendix records the search so that a reader can judge it or extend it.

The question. Is it stated or proved anywhere that rowmotion on a minuscule poset, carried to the weights by the natural bijection, is or is not itself an element of the Weyl group; for which minuscule posets it or its half-turn is; that a pair of weights keeps its inner product under one step of rowmotion according to a parity of toggle data; or what fraction of pairs it keeps?

Searches 1–3 (September 2026, both sides; keyword). Toggle groups, periodicity, homomesy, cyclic sieving; rowmotion and an ambient inner-product sign on minuscule weights; rowmotion with quadratic or bilinear forms mod 2; the E_7 56-ideal case and the Gosset graph. Found: the Coxeter-motion and birational literature (Okada 2021), the toggle literature (Striker–Williams 2012; Striker 2016), the homomesy literature (Propp–Roby 2015; Rush–Wang 2015; Defant–Hopkins–Poznanović–Propp 2021). No match.

Search 4 (Balashov, 2026-09-10; citation trail). The thirty documents citing Rush–Shi (2013) on zbMATH Open, all read. No match; the two nearest (Marczinik–Thomas–Yıldırım 2024; Panyushev 2016) are discussed in §7.4.

Search 5 (Stenberg side, 2026-09-19; corpus). The arXiv API query `all:rowmotion` (title, abstract and full-text index), 200 results requested, returned the complete corpus of **57 documents** (2014–2026). Every title and abstract was read. Titles containing *minuscule*: Hopkins 2016 (CDE property for minuscule lattices), Hopkins 2019 (minuscule doppelgängers), Okada 2020 (birational rowmotion and Coxeter-motion), Pechenik 2021 (minuscule analogues of the Cameron–Fon-Der-Flaass conjecture). Titles containing *Coxeter*: Okada 2020, Marczinik–Thomas–Yıldırım 2022. Titles containing *Weyl*, *isometry*, *orthogonal*, *inner product* or E_7 : none. The remaining titles concern Tamari and Cambrian lattices, fences, root posets and rowvacuation, interval-closed sets, products of chains, birational and piecewise-linear lifts, Markov chains, rooted trees, trapezoids, plane partitions, semidistributive and trim lattices, alternating sign matrices, and echelonmotion. The same query to the zbMATH Open API returned nine documents (Einstein–Propp 2014, 2021; Grinberg–Roby 2015, 2016; Striker–Williams 2012; Bernstein–Striker–Vorland 2021, 2024; Defant–Hopkins–Poznanović–Propp 2023; Propp–Roby 2015), none with the terms above in title or keywords. (The corpus query `all:rowmotion AND all:minuscule`, sorted by date, returned Bernstein–Striker–Vorland 2022, Pechenik 2021, Okada 2020, Hopkins 2019, Vorland 2017, Hopkins 2016.)

Search 6 (Stenberg side, 2026-09-19; the primary texts). Read for the specific question:

- *Rush–Shi (2013)*. Theorem 1.3: rowmotion and the toggle product $t_{(i_1, \dots, i_n)}$ are conjugate in the toggle group; Theorem 1.4: under Stembridge’s bijection $J(P) \rightarrow W^J$, that toggle product corresponds to a Coxeter element c . They do not say that rowmotion itself is or is not a Coxeter element’s action; the chain case is attributed to Stanley (Striker–Williams 2012, Thm 6.1); the D_n vector case receives no separate treatment; inner products, isometries and pairs of weights are not mentioned.
- *Okada (2021)*. Coxeter-motion is defined as a product of file toggles (his eq. (10)) and shown conjugate to rowmotion in the birational toggle group (Thm 4.3): two maps, conjugate, not equal. The double-tailed diamond ($D_n \omega_1$) is treated in his §5.2 with separate formulas for the two tails — the birational shadow of the same poset on which Theorem 2 finds the half-turn linear — without any statement about Weyl group membership.

- *Hopkins (2020)*. Records “the action of rowmotion is conjugate to the action of a Coxeter element” (§5.2.1) and the open conjectures on cyclic sieving (Conj. 5.8) and doppelgänger (Conj. 5.3); nothing on Weyl membership, isometry or pairs. Cites Rush–Shi, Grinberg–Roby, Garver–Patrias–Thomas 2023 and Okada on minuscule rowmotion.
- *Thomas–Williams (2019)*. Trim lattices and rowmotion in slow motion; no statement about Weyl group membership or the weights.

Search 7 (Stenberg side, 2026-09-19; phrasing). Web searches for rowmotion with “Weyl group element”, “isometry”, “signed permutation”, “vector representation”, “half-spin”, “ E_7 ”, “cyclic shift” and “Coxeter element ... coincide” returned only the documents above.

What the search does not reach. MathSciNet and the full text of zbMATH’s review database were not available; Google Scholar was reached only through a general web index; work in progress and theses are not indexed by the arXiv query. A reader with MathSciNet who finds the statements of §§7.2, 7.6, 7.7 as known corollaries is asked to say so; the paper’s claims are worded to survive that.